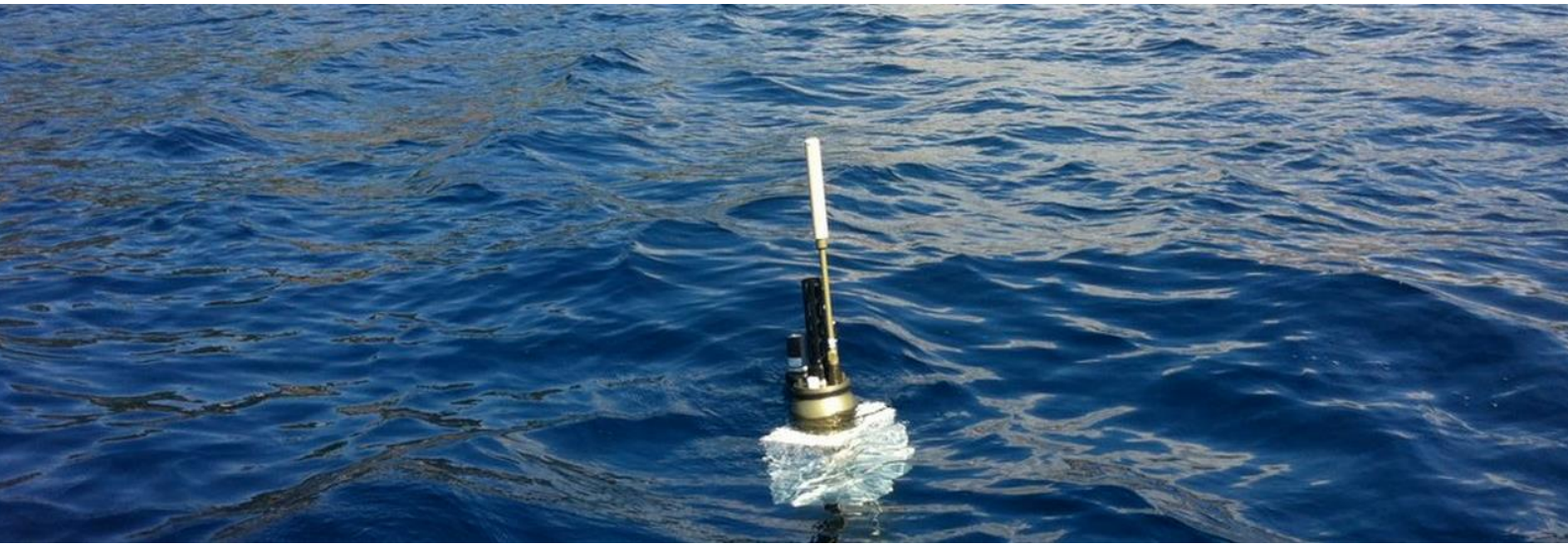
PFV2 Profiler - Parameter set

PROFILING **F**LOAT **V**ECTOR **2**



NTxxxxxx_PFV2_Parameter_Set_rev1.docx
Revision 1 (2024-08-26)

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1. Revision history

Revision	Release date	Notes	Author
1.0	2024-04-15	Original	D. NOGRE
	2024-07-11	Setting <ALARMS-END_OF_LIFE-P11> deleted	D. NOGRE
	2024-08-21	Setting <TECHNICAL-HYDRAULIC-P2> added	D. NOGRE
	2024-08-26	Corrections from JP Rannou	

2. Parameter setting

2.1 Configuration file

Parameter setting of the float is performed using a configuration file "setting_update.xml". This file is organized in nodes. Each section contains keys from P0 to Px.

2.2 List of parameters

Each parameter is controlled by an access right:

- "S" : Standard – Access for all users
- "E" : Expert – Access is protected by the expert password
- "A" : Admi – Access is also protected but by the administrator password

<MISSION>
<SETTING>
<MISSION …>

No.	Comment	Limitation	Access rights
0	Reference day of the 1st cycle (days)	[0-40]	S

<PRESSURE_DETECTION>
<SETTING>
<MISSION>
<PRESSURE_DETECTION>

No.	Comment	Limitation	Access rights
0	Period of pressure monitoring (s)	[30-1800(30mn)]	A
1	Threshold of pressure to start mission (dbar)	[1.0-100.0]	A
2	Maximum number of days before starting mission (days)	[1-365]	A

<INITIALIZATION>
<SETTING>
<MISSION>
<INITIALIZATION …>

No.	Comment	Limitation	Access rights
0	Waiting at surface before starting mission (s)	[0-86400]	S

<LOOPxx>
<SETTING>
<MISSION>
<LOOPxx …>

No.	Comment	Limitation	Access rights
0	Repetition's number of this same loop	[0-65535]	S

<CYCLExx>
<SETTING>
<MISSION>
<LOOPxx>
<CYCLExx …>

No.	Comment	Limitation	Access rights
0	Number of cycles	[0-65535]	S
1	Period of cycle (s)	[0-1728000(480h)]	S
2	Expected hour at surface	hh:mm:ss	S

<SETTING>
<MISSION>
<LOOPxx>
<CYCLExx>
<TRANSMISSION01 …>

No.	Comment	Limitation	Access rights
0	Enable/disable transmission session	[True/False]	S

<SETTING>
<MISSION>
<LOOPxx>
<CYCLExx>
<BUOYANCY_RED …>

No.	Comment	Limitation	Access rights
0	Mode of buoyancy reduction 0: SECURE 1: FAST	[0-1]	E

<SETTING>
<MISSION>
<LOOPxx>
<CYCLExx>
<DESC_PARK …>

No.	Comment	Limitation	Access rights
0	Expected depth (dbar)	[0.0-2020.0]	S
1	Mean speed on descent (cm/s)	[0.5-100.0]	E

<SETTING>
<MISSION>
<LOOPxx>
<CYCLExx>

<PARK_DRIFT …>

No.	Comment	Limitation	Access rights
0	Minimum duration in drift (s)	[0-1728000(480h)]	S
1	Tolerance of positioning (dbar)	[0.0-200.0]	E
2	Tolerance of repositioning (dbar)	[0.0-100.0]	E

<SETTING>
<MISSION>
<LOOPxx>
<CYCLExx>
<DESC_PROF …>

No.	Comment	Limitation	Access rights
0	Expected depth (dbar)	[0.0-2020.0]	S
1	Mean speed on descent (cm/s)	[0.5-100.0]	E

<SETTING>
<MISSION>
<LOOPxx>
<CYCLExx>
<PROF_DRIFT …>

No.	Comment	Limitation	Access rights
0	Minimum duration in drift (s)	[0-1728000(480h)]	S
1	Tolerance of positioning (dbar)	[0.0-200.0]	E
2	Tolerance of repositioning (dbar)	[0.0-100.0]	E

<SETTING>
<MISSION>
<LOOPxx>
<CYCLExx>
<ASCENT …>

No.	Comment	Limitation	Access rights
0	Expected depth (0 if only 1 ascent phase)	[0.0-2020.0]	S
1	Mean speed on ascent (cm/s)	[0.5-100.0]	E

<SETTING>
<MISSION>
<LOOPxx>
<CYCLExx>
<IN_AIR …>

No.	Comment	Limitation	Access rights
0	Cycle periodicity (1/x where x=number of the cycle)	[0-250]	S
1	Duration of waiting at surface (s)	[0-3600(1h)]	S



<SETTING>
<MISSION>
<LOOPxx>
<CYCLExx>
<END_OF_CYCLE …>

No.	Comment	Limitation	Access rights
0	Waiting at the end of the cycle (s)	[0-21600(6h)]	S

<END_OF_LIFE>
<SETTING>
<MISSION>
<END_OF_LIFE …>

No.	Comment	Limitation	Access rights
0	Periodicity of the GPS/transmission session (s)	[0-432000(120h)]	S

<TECHNICAL-BUOYANCY_RED>
<SETTING>
<TECHNICAL>
<BUOYANCY_RED …>

No.	Comment	Limitation	Access rights
0	1st threshold of the buoyancy reduction (dbar)	[1.0-20.0]	E
1	2nd threshold of the buoyancy reduction (dbar)	[1.0-25.0]	E
2	Period of pressure monitoring (s)	[10-3600(1h)]	E
3	Volume of SV activation (cm3)	[1.0-50.0]	E

<TECHNICAL-DESCENT>
<SETTING>
<TECHNICAL>
<DESCENT …>

No.	Comment	Limitation	Access rights
0	Tolerance in pressure of expected depth (dbar)	[0.0-200.0]	E

<SETTING>
<TECHNICAL>
<DESCENT>
<MIN …> / <MAX …>

No.	Comment	Limitation	Access rights
0	Mean speed (cm/s)	[0.5-100.0]	A
1	Min speed (cm/s)	[0.5-100.0]	A
2	Period of pressure monitoring (s)	[10-3600]	A
3	Volume of SV activation (cm3)	[1.0-50.0]	A

<TECHNICAL-DRIFT>
<SETTING>
<TECHNICAL>
<DRIFT …>

No.	Comment	Limitation	Access rights
0	Period of pressure monitoring (s)	[60-86400]	E

<TECHNICAL-REPOSITIONING>
<SETTING>
<TECHNICAL>
<REPOSITIONING …>

No.	Comment	Limitation	Access rights
0	Speed stabilization (cm/s)	[0.1-100.0]	A

<SETTING>
<TECHNICAL>
<REPOSITIONING>
<MIN …> / <MAX …>

No.	Comment	Limitation	Access rights
0	Tolerance in pressure of expected depth (dbar)	[0.5-100.0]	A
1	Period of pressure monitoring (s)	[10-3600]	A
2	Maximum volume of SV activation (cm3)	[1.0-50.0]	A
3	Maximum volume of PUMP activation (cm3)	[1.0-50.0]	A

<TECHNICAL-ASCENT>
<SETTING>
<TECHNICAL>
<ASCENT …>

No.	Comment	Limitation	Access rights
0	Tolerance in pressure of expected depth (dbar)	[0.0-100.0]	E

<SETTING>
<TECHNICAL>
<ASCENT>
<SURFACE …>

No.	Comment	Limitation	Access rights
0	Threshold of pressure considering that the surface has been reached (dbar)	[3.0-50.0]	E
1	Waiting delay once surface reached (s)	[0-3600]	E

<SETTING>
<TECHNICAL>
<ASCENT>
<MIN …> / <MAX …>

No.	Comment	Limitation	Access rights
0	Mean speed (cm/s)	[0.5-100.0]	A
1	Min speed (cm/s)	[0.5-100.0]	A
2	Period of pressure monitoring (s)	[10-3600]	A
3	Volume of PUMP activation (cm3)	[1.0-50.0]	A

<TECHNICAL-EMERGENCE>
<SETTING>
<TECHNICAL>

<EMERGENCE …>

No.	Comment	Limitation	Access rights
0	Volume of pumping increasing buoyancy	[200.0-800.0]	E

<TECHNICAL-IN_AIR>
<SETTING>
<TECHNICAL>
<IN_AIR …>

No.	Comment	Limitation	Access rights
0	Volume of pumping increasing buoyancy to make acquisition in air	[0.0-800.0]	E

<TECHNICAL-TRANSMISSION>
<SETTING>
<TECHNICAL>
<TRANSMISSION …>

No.	Comment	Limitation	Access rights
0	GPS: Time-out of session (s)	[300-21600]	E
1	GPS: Volume of pumping in case of acquisition failure (cm3)	[50.0-800.0]	E
2	IRIDIUM: Time-out of 1st connection (s)	[300-21600]	E
3	IRIDIUM: Time-out of session (s)	[300-86400]	E
4	Enable/disable GZ compression on all files sent	[True/False]	E
5	Number of sending tries before quarantining a file	[5-100]	E

<SETTING>
<TECHNICAL>
<TRANSMISSION>
<RUDICS …> ONLY FOR RUDICS VECTOR

No.	Comment	Limitation	Access rights
0	Maximum size of transmit files (KB)	[1-20]	A
1	Reset RUDICS connection before retrieving remote control commands	[True/False]	A

<SETTING>
<TECHNICAL>
<TRANSMISSION>
<SBD …> ONLY FOR SBD VECTOR

No.	Comment	Limitation	Access rights

0	Resend setting file one remote control commands receiving	[True/False]	E

<TECHNICAL-EMERGENCY_ASCENT>
<SETTING>
<TECHNICAL>
<EMERGENCY_ASCENT …>

No.	Comment	Limitation	Access rights
0	Min speed (cm/s)	[0.5-100.0]	E
1	Volume of PUMP activation (cm3)	[1.0-50.0]	E

<TECHNICAL-LOGGER>
<SETTING>
<TECHNICAL>
<LOGGER …>

No.	Comment	Limitation	Access rights
0	Event logger level 0: Project information 1: Technical information 2: Endurance information	[0-2]	E
1	Pressure information Bit 0: On descent Bit 1: On drift Bit 2: On ascent	b00000xxx	E

<TECHNICAL-SECURITY>
<SETTING>
<TECHNICAL>
<SECURITY …>

No.	Comment	Limitation	Access rights
0	Surface pressure (dbar)	[1.0-50.0]	E
1	Min delay between pressure to compute speed (s)	[10-300]	A
2	Volume of PUMP activation on descent braking (cm3)	[1.0-50.0]	E
3	Volume of PUMP activation on emergency braking (cm3)	[1.0-50.0]	E

<SETTING>
<TECHNICAL>
<SECURITY>
<GROUNDING …>

No.	Comment	Limitation	Access rights
0	Volume of SV at surface before considering a grounding (cm3)	[1.0-1600.0]	E
1	Volume of SV on descent before considering a grounding (cm3)	[10.0-200.0]	E

2	Speed of stabilization on descent (cm/s)	[0.1-100.0]	A
3	Behaviour on grounding in “descent to parking” or “parking drift” 0: Pressure switch 1: Stay grounded 2: Immediate ascent	[0-2]	E
4	Behaviour on grounding in “descent to profile” or “profile drift” 0: Pressure switch 1: Stay grounded 2: Immediate ascent	[0-2]	E
5	Minimum pressure before allowing switch on depth setpoint (dbar)	[50.0-1000.0]	E
6	Offset pressure of depth setpoint (dbar)	[0.0-200.0]	E

<TECHNICAL-HYDRAULIC>

<SETTING>

<TECHNICAL>

<HYDRAULIC …>

No.	Comment	Limitation	Access rights
0	Minimum duration of SV activation (ms)	[10.0-50.0]	A
1	Minimum duration of PUMP activation (ms)	[10.0-50.0]	A
2	Multiplier coefficient of hydraulic algorithm	[0.1-5.0]	A

<SENSORS-SENSORxx>

<SETTING>

<SENSORS>

<SENSORxx …>

No.	Comment	Limitation	Access rights
0	Enable/disable the sensor	[True/False]	S
1	Format of treated data	Value depending on each sensor	E
2	Datation frequency of treated data	[0-65535]	E

<SETTING>

<SENSORS>

<SENSORxx>

<DESCENT …>

No.	Comment	Limitation	Access rights
0	Enable/disable acquisition on “descent to parking”	[True/False]	S
1	Enable/disable acquisition on “descent to profile”	[True/False]	S

<SETTING>

<SENSORS>

<SENSORxx>

<ASCENT …>

No.	Comment	Limitation	Access rights
0	Enable/disable acquisition on “ascent”	[True/False]	S

<SETTING>
<SENSORS>
<SENSORxx>
<DESCENT> / <ASCENT>
<ZONE1 …>

No.	Comment	Limitation	Access rights
0	Zone 1 - Pressure threshold between zone 1 and zone 2	[0.1-2500.0]	S
1	Zone 1 - Sampling period (s)	[0.0-864000(240h)]	S
2	Zone 1 - Acquisition type 0 : None 1 : Continu 2 : Pulsed 3 : Spot sampling	[0-3]	S
3	Zone 1 - Processing type 00 : Raw 80 : Raw decimated 81 : Average decimated	[00/80/81]	S
4	Zone 1 - Slice thickness (dbar)	[0.01-1000.0]	S
5	Zone 1 - Number of samples in spot sampling mode	[1-65535]	S

<SETTING>
<SENSORS>
<SENSORxx>
<DESCENT> / <ASCENT>
<ZONE2 …>

No.	Comment	Limitation	Access rights
0	Zone 2 - Pressure threshold between zone 2 and zone 3	[0.1-2500.0]	S
1	Zone 2 - Sampling period (s)	[0.0-864000(240h)]	S
2	Zone 2 - Acquisition type 0 : None 1 : Continu 2 : Pulsed 3 : Spot sampling	[0-3]	S
3	Zone 2 - Processing type 00 : Raw 80 : Raw decimated 81 : Average decimated	[00/80/81]	S
4	Zone 2 - Slice thickness (dbar)	[0.01-1000.0]	S
5	Zone 2 - Number of samples in spot sampling mode	[1-65535]	S

<SETTING>

<SENSORS>

<SENSORxx>

<DESCENT> / <ASCENT>

<ZONE3 …>

No.	Comment	Limitation	Access rights
0	Zone 3 - Pressure threshold between zone 3 and zone 4	[0.1-2500.0]	S
1	Zone 3 - Sampling period (s)	[0.0-864000(240h)]	S
2	Zone 3 - Acquisition type 0 : None 1 : Continu 2 : Pulsed 3 : Spot sampling	[0-3]	S
3	Zone 3 - Processing type 00 : Raw 80 : Raw decimated 81 : Average decimated	[00/80/81]	S
4	Zone 3 - Slice thickness (dbar)	[0.01-1000.0]	S
5	Zone 3 - Number of samples in spot sampling mode	[1-65535]	S

<SETTING>

<SENSORS>

<SENSORxx>

<DESCENT> / <ASCENT>

<ZONE4 …>

No.	Comment	Limitation	Access rights
0	Zone 4 - Pressure threshold between zone 4 and zone 5	[0.1-2500.0]	S
1	Zone 4 - Sampling period (s)	[0.0-864000(240h)]	S
2	Zone 4 - Acquisition type 0 : None 1 : Continu 2 : Pulsed 3 : Spot sampling	[0-3]	S
3	Zone 4 - Processing type 00 : Raw 80 : Raw decimated 81 : Average decimated	[00/80/81]	S
4	Zone 4 - Slice thickness (dbar)	[0.01-1000.0]	S
5	Zone 4 - Number of samples in spot sampling mode	[1-65535]	S

<SETTING>

<SENSORS>

<SENSORxx>

<DESCENT> / <ASCENT>

<ZONE5 …>

No.	Comment	Limitation	Access

			rights
1	Zone 5 - Sampling period (s)	[0.0-864000(240h)]	S
2	Zone 5 - Acquisition type 0 : None 1 : Continu 2 : Pulsed 3 : Spot sampling	[0-3]	S
3	Zone 5 - Processing type 00 : Raw 80 : Raw decimated 81 : Average decimated	[00/80/81]	S
4	Zone 5 - Slice thickness (dbar)	[0.01-1000.0]	S
5	Zone 5 - Number of samples in spot sampling mode	[1-65535]	S

<SETTING>
<SENSORS>
<SENSORxx>
<PARK_DRIFT>

No.	Comment	Limitation	Access rights
0	Sampling period (s)	[0.0-864000(240h)]	S

<SETTING>
<SENSORS>
<SENSORxx>
<PROF_DRIFT>

No.	Comment	Limitation	Access rights
0	Sampling period (s)	[0.0-864000(240h)]	S

<SETTING>
<SENSORS>
<SENSORxx>
<IN_AIR>

No.	Comment	Limitation	Access rights
0	Sampling period (s)	[0.0-3600.0(1h)]	S
1	Acquisition type 0 : None 1 : Continu 2 : Pulsed	[0-2]	S

<SETTING>
<SENSORS>
<SENSORxx>

<TECHNICAL >

No.	Comment	Limitation	Access rights
0	Warmup delay (s)	Depends on sensor	A
1	Power-off delay (s)	Depends on sensor	A
2	Theorical delay between 2 samples (s)	Depends on sensor	E
3	Index of 1st sample valid	Depends on sensor	E

<SETTING>
<SENSORS>
<SENSOR>
<SPECIFIC > Parameter setting specific to SBE41 sensor.

No.	Comment	Limitation	Access rights
0	"Cut-off" pressure (dbar)	[0.0-100.0]	E

<ICE>
<SETTING>
<ICE>

No.	Comment	Limitation	Access rights
0	Key validating the option activation	32bits	E
1	Enable/disable ice detection	[True/False]	S
2	Maximum days without trying a transmission session (days)	[0-365]	S
3	Maximum days before forcing a transmission session (days)	[0-365]	S
4	Depth for "slow down" speed of ascent (dbar)	[0.0-2500.0]	S
5	“Slow down” speed (cm/s)	[0.5-100.0]	S

<SETTING>
<ICE>
<ISA>

No.	Comment	Limitation	Access rights
0	Enable/disable ISA detection	[True/False]	S
1	Collection starting depth (dbar)	[0.0-300.0]	S
2	Collection stopping depth (dbar)	[0.0-300.0]	S
3	Median temperature threshold (°C)	[(-100.0)-100.0]	S
4	Successive detections counter threshold	[0-255]	S

<SETTING>
<ICE>
<HANGING>

No.	Comment	Limitation	Access rights
0	Enable/disable hanging detection	[True/False]	S

1	Maximum volume before considering a hanging (cm3)	[100.0-850.0]	E
2	Speed stabilization (cm/s)	[0.1-100.0]	E
3	Pressure delta to flow back (dbar)	[0.0-50.0]	E

<SETTING>

<ICE>

<SAT_MASK>

No.	Comment	Limitation	Access rights
0	Enable/disable satellite mask detection	[True/False]	S

<SELFTEST>

<SETTING>

<SELFTEST>

No.	Comment	Limitation	Access rights
0	Minimum voltage of battery (V)	[4.0-18.0]	A
1	Minimum external pressure (dbar)	[(-100.0)-2500.0]	A
2	Maximum external pressure (dbar)	[100.0-2500.0]	A
3	Minimum internal pressure (mbar)	[10.0-2000.0]	A
4	Maximum internal pressure (mbar)	[10.0-2000.0]	A
5	Minimum internal temperature (°C)	[(-40.0)-85.0]	A
6	Maximum internal temperature (°C)	[(-40.0)-85.0]	A
7	Minimum internal relative humidity (%RH)	[0.0-100.0]	A
8	Maximum internal relative humidity (%RH)	[0.0-100.0]	A

<ALARMS>

<SETTING>

<ALARMS>

<THRESHOLD>

No.	Comment	Limitation	Access rights
0	Minimum voltage of battery (V)	[4.0-18.0]	E
1	Minimum external pressure (dbar)	[(-100.0)-2500.0]	E
2	Maximum external pressure (dbar)	[100.0-2500.0]	E
3	Minimum internal pressure (mbar)	[10.0-2000.0]	E
4	Maximum internal pressure (mbar)	[10.0-2000.0]	E
5	Minimum internal temperature (°C)	[(-40)-85.0]	E
6	Maximum internal temperature (°C)	[(-40)-85.0]	E
7	Minimum internal relative humidity (%RH)	[0.0-100.0]	E
8	Maximum internal relative humidity (%RH)	[0.0-100.0]	E
9	Gap between 2 consecutive pressure requests (dbar)	[0.0-50.0]	E
10	Maximum speed on descent (cm/s)	[0.0-1000.0]	E

<SETTING>

<ALARMS>

<END_OF_LIFE>

No.	Comment	Limitation	Access rights
0	Detection of minimum battery voltage	[True/False]	E
2	Detection of maximum external pressure	[True/False]	E
4	Detection of maximum internal pressure	[True/False]	E
10	Detection of maximum speed on descent	[True/False]	E
~~11~~	~~Detection of failure on external pressure sensor~~	~~[True/False]~~	~~E~~
21	Detection of insulation voltage	[True/False]	E

3. Float installation

3.1.Ballasting

The float can be deployed with the bladder either empty or full. The float automatically inflates its bladder before the mission. In that case, the float is in "floating" deployment.

3.2.Self-test

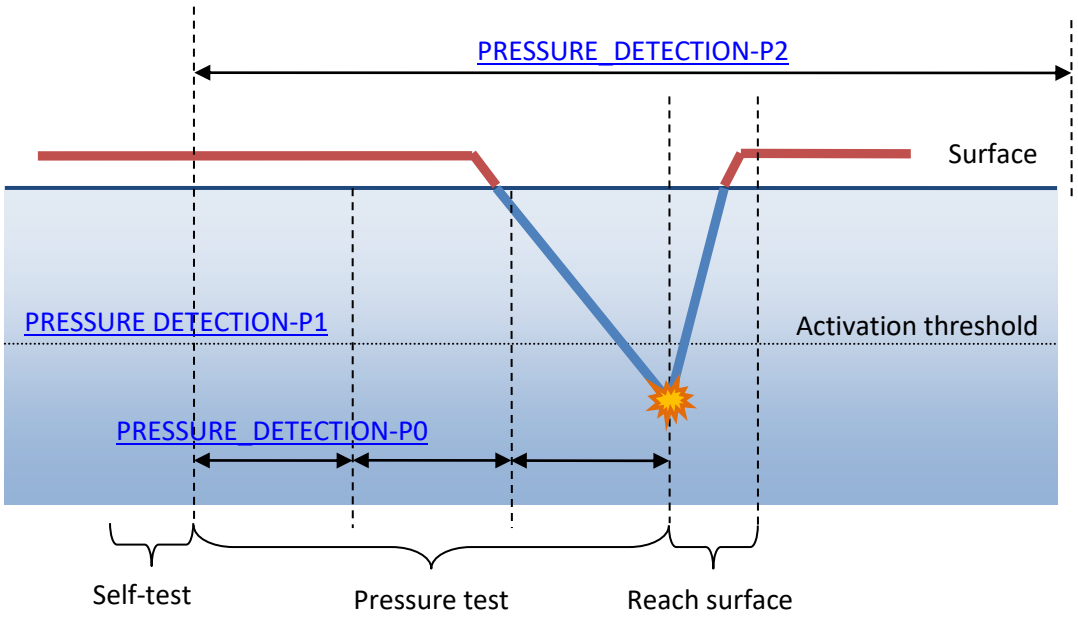
Before the mission starts, the float performs a system self-test. His result is sent by Iridium link.

3.3."Floating" deployment of float

3.3.1. "Pressure detection" deployment

For pressure detection to work, the float must be in a “sinking” condition. It must be able to sink from the surface to the pressure activation threshold when deployed.

The pressure activation starts with a successful self-test. The float enters a pressure test phase during which it monitors the pressure periodically. When the pressure exceeds the detection threshold, the float increases its buoyancy to reach the surface and starts the mission.



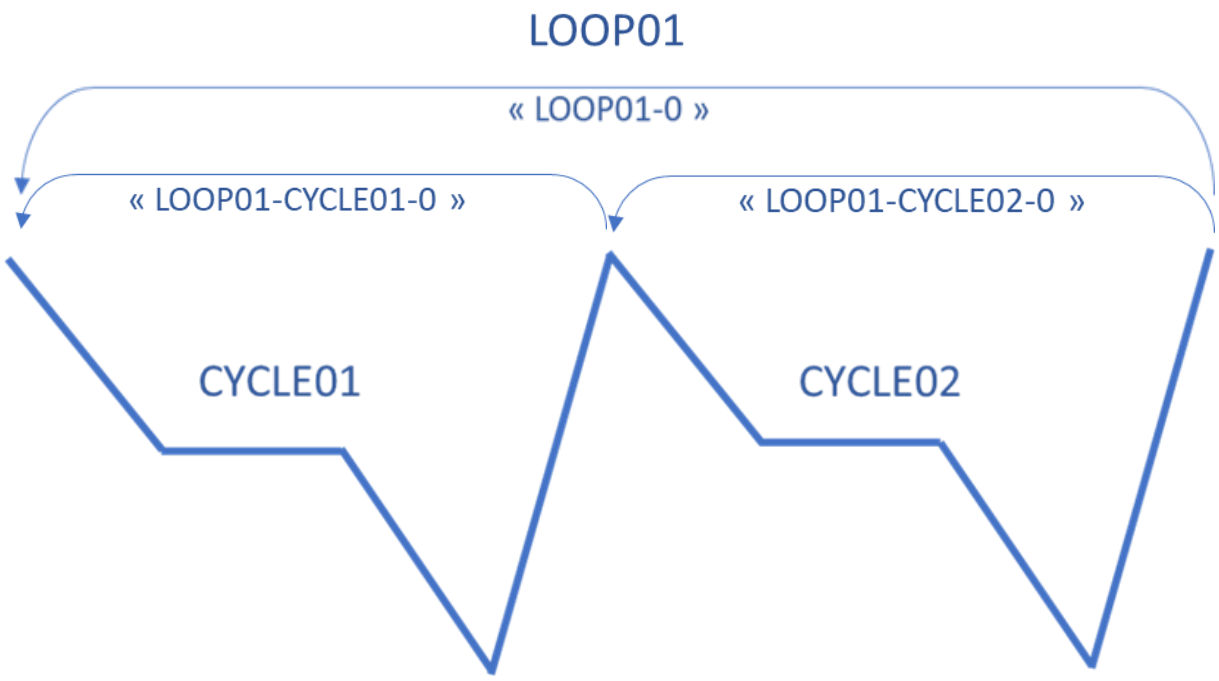
In case the pressure test phase fails, the float will engage the mission after a maximum duration.

When the mission starts (end of pressure test succeed or fails) the arming state sets to “pressure detection” is switched to “standard” state automatically.

4. Setting a mission

To define the mission, the configuration introduces the notions of "LOOP" and "CYCLE".

A loop includes one or more cycles. The interest of this loop is that the different pattern of cycles can be repeated. And by different sets of parameters, the float can perform a wide range of cycling.



4.1. "LOOP" meaning

As its name suggests, a loop is a function that repeats one or more cycles.

- LOOPxx-P0: Defined the number of "LOOP" repeats.

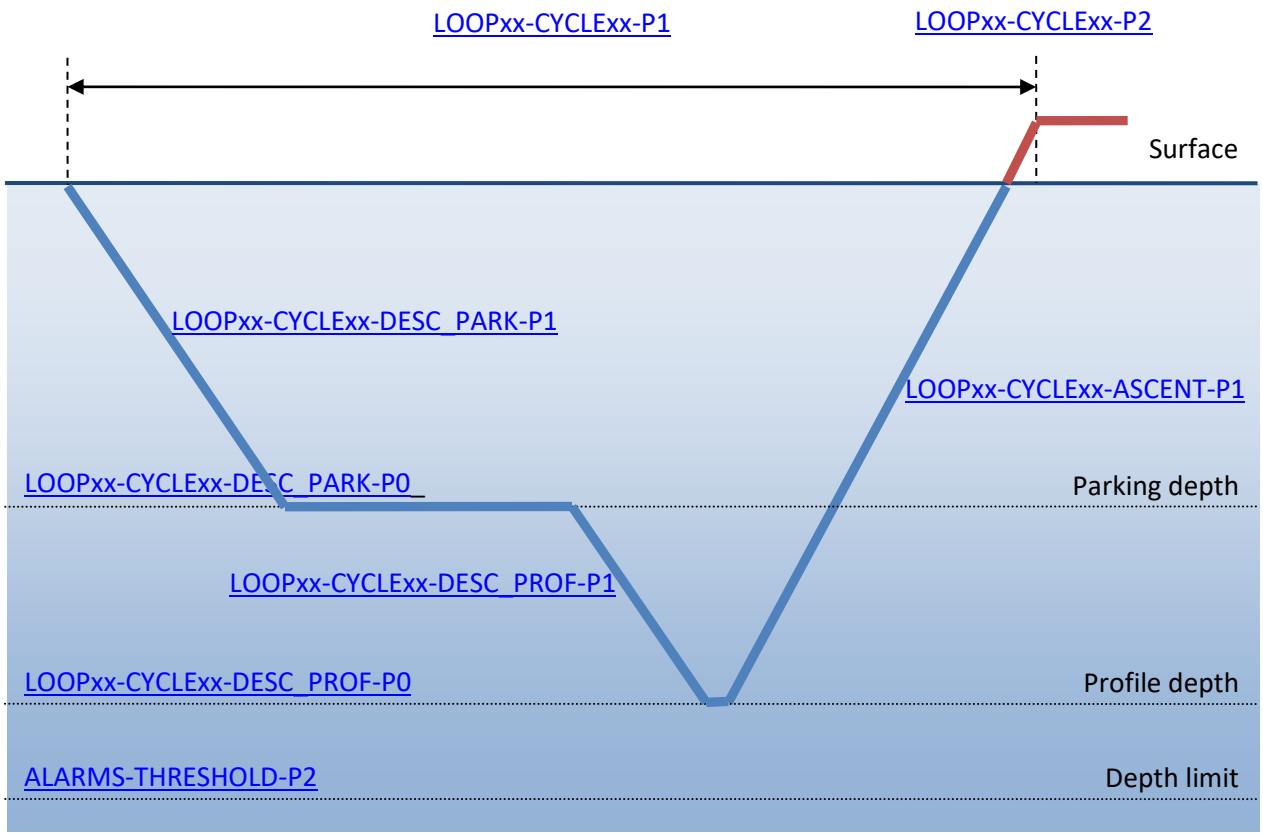
4.2. "CYCLE" meaning

A cycle is defined as the execution of a profile followed by a session at the surface (GPS, transmission...)

- LOOPxx-CYCLExx-P0: Number of repeats of the same cycle before switching to the next.

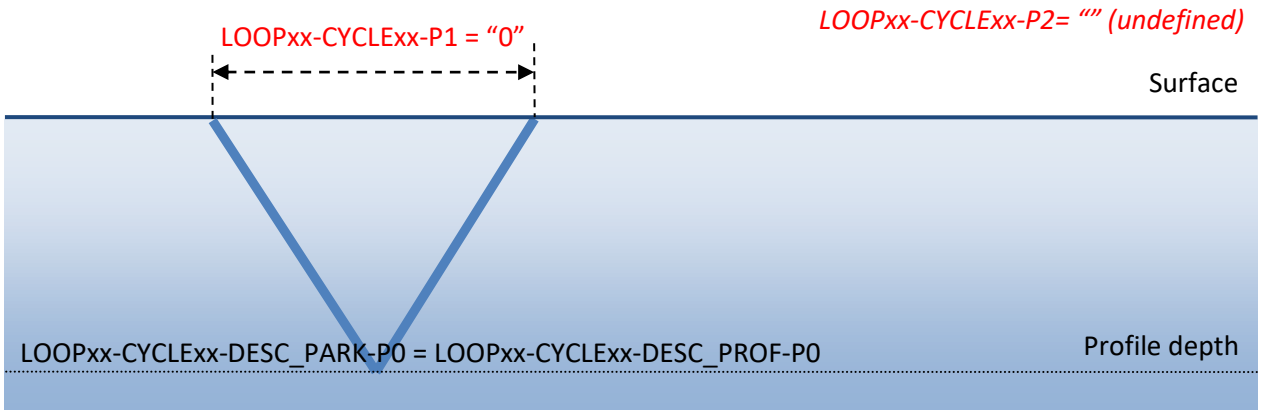
Its operation is based on 2 parameters:

- LOOPxx-CYCLExx-P1: Cycle period
- LOOPxx-CYCLExx-P2: Expected hour at surface



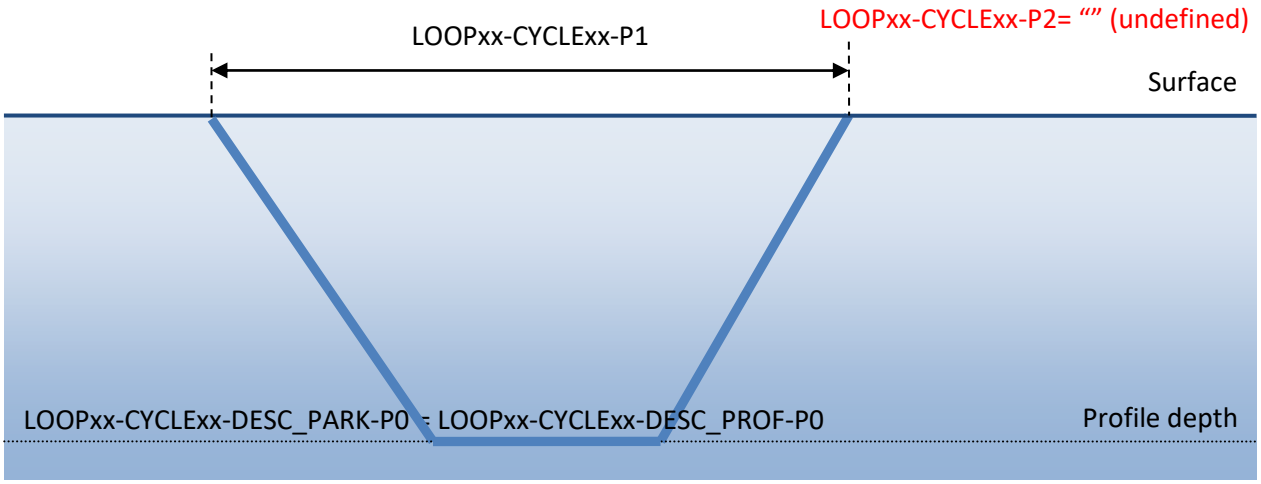
4.3.Case 1: maintaining the profile depth

The profile is performed to observe the profile depth. Drift phases are not considered.



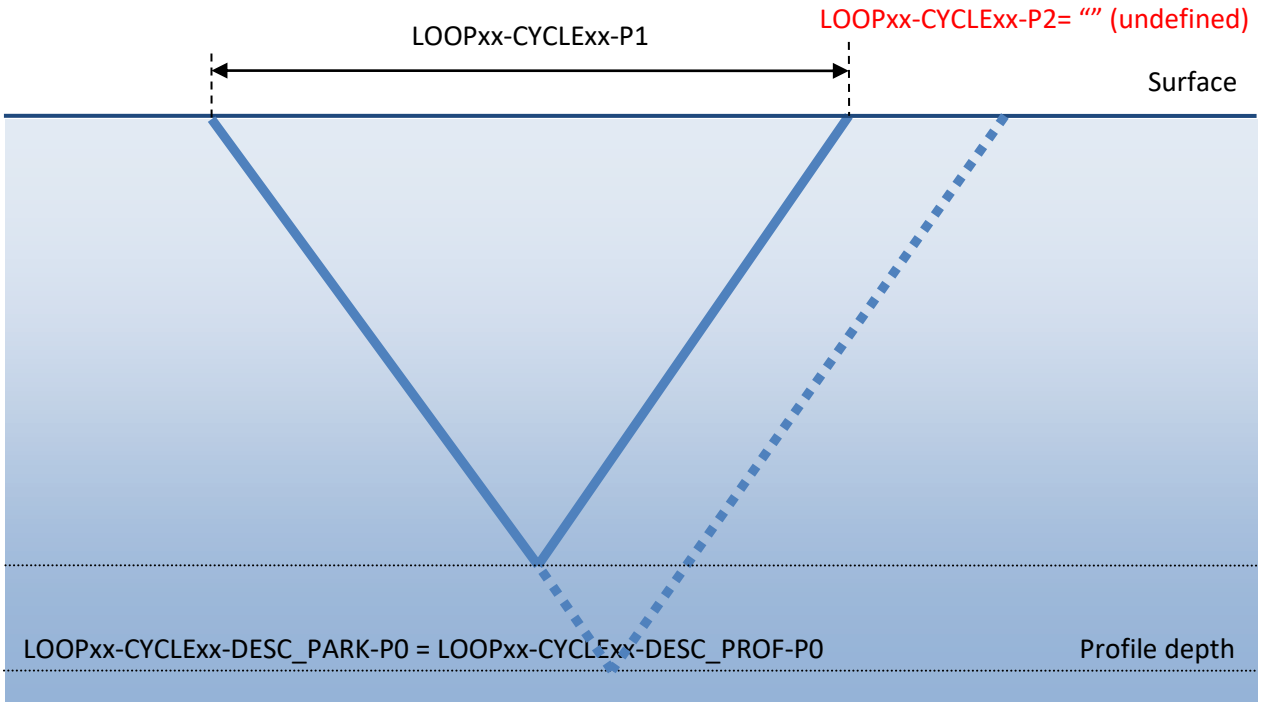
4.4.Case 2: maintaining the profile duration (depth reached)

The profile is performed with standby at parking depth to observe the profile duration.



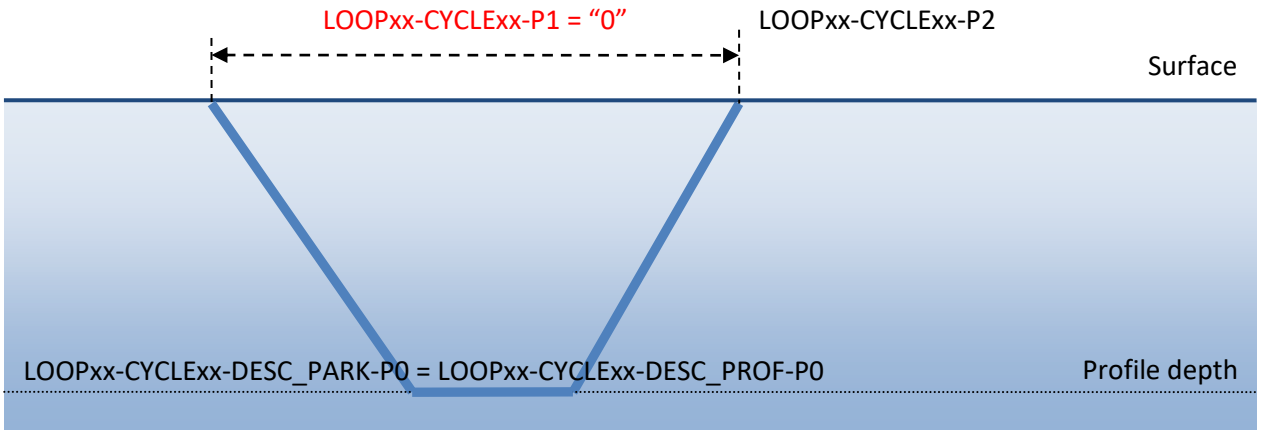
4.5.Case 3: maintaining the profile duration (depth not reached)

If the profile cannot be performed within the time allocated, the ascent is anticipated to observe the profile duration.

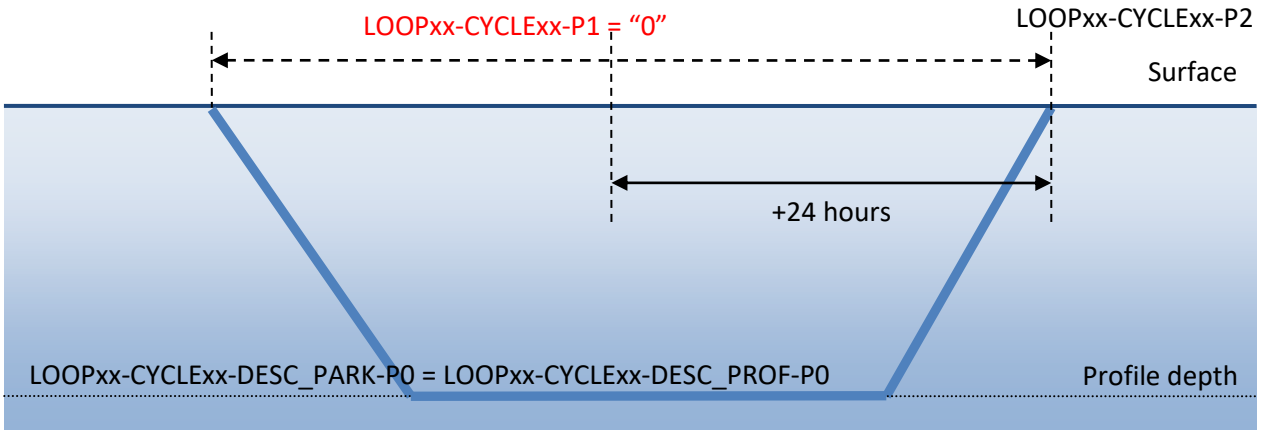


4.6.Case 4: maintaining measurement depth and surface synchronization

The profile is performed with standby at parking depth to observe the time of presence at the surface and the measurement depth.

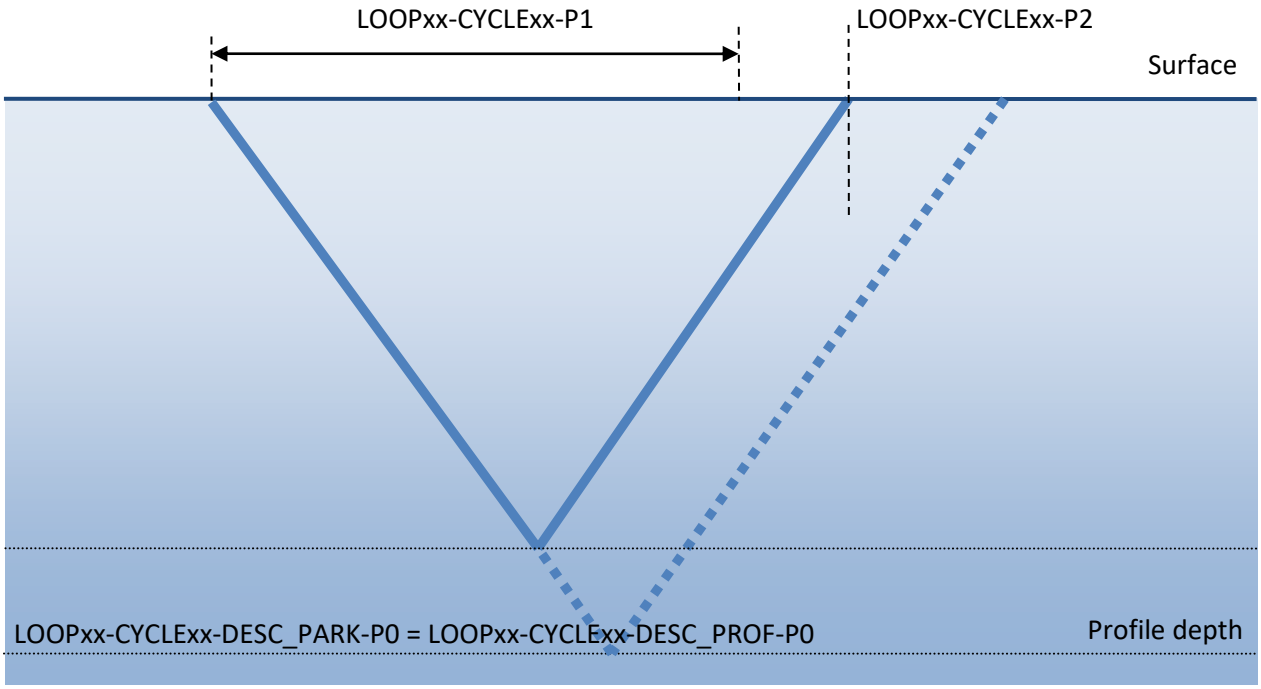


If the time of presence at the surface is already past, the time of presence at the surface is postponed to the following day.



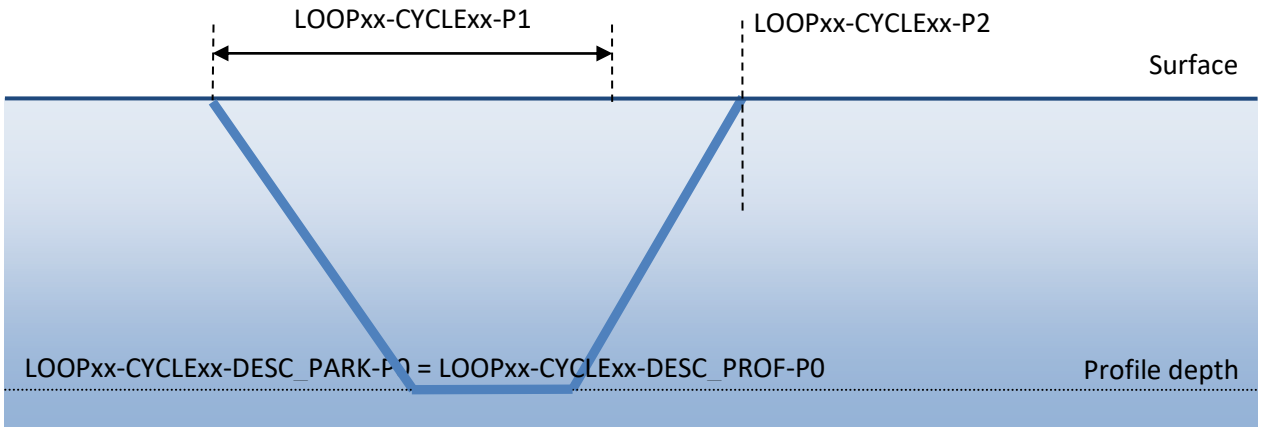
4.7.Case 5: maintaining surface synchronization (depth not reached)

If the profile cannot be performed within the time allocated, the ascent is anticipated to observe the time of presence at the surface.



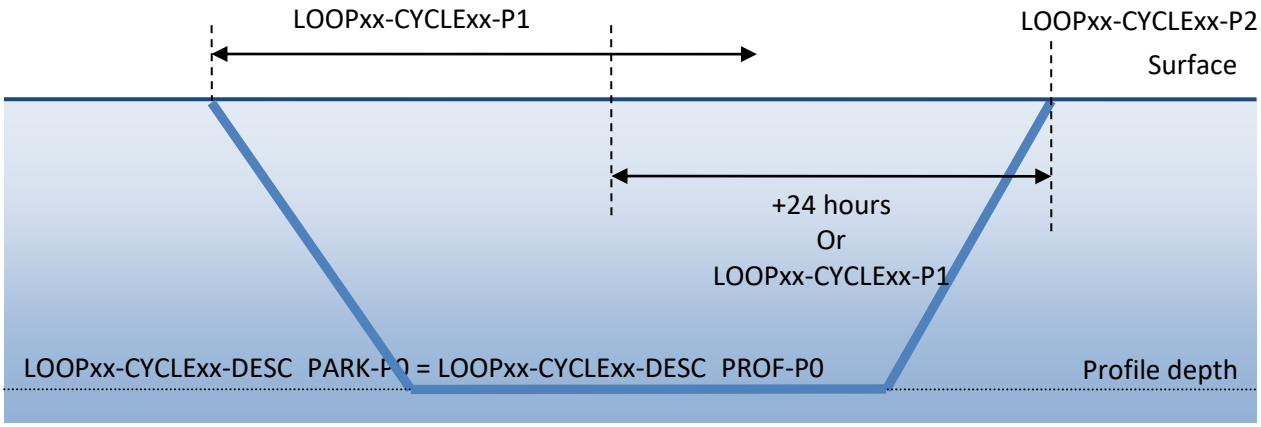
4.8.Case 6: maintaining surface synchronisation (depth reached)

The profile is performed with standby at parking depth to observe the time of presence at the surface and the minimum duration.



If the profile cannot be performed while observing the time of presence at the surface and the minimum duration, the time of presence at the surface is postponed to:

- 24 hours if “LOOPxx-CYCLExx-P1” is modulo 24 hours
- “LOOPxx-CYCLExx-P1” if “LOOPxx-CYCLExx-P1” is not modulo 24 hours



5. Surface management

5.1.Emergence reduction

The emergence reduction consists in decreasing the buoyancy of the float following a session at the surface, until sinking is detected.

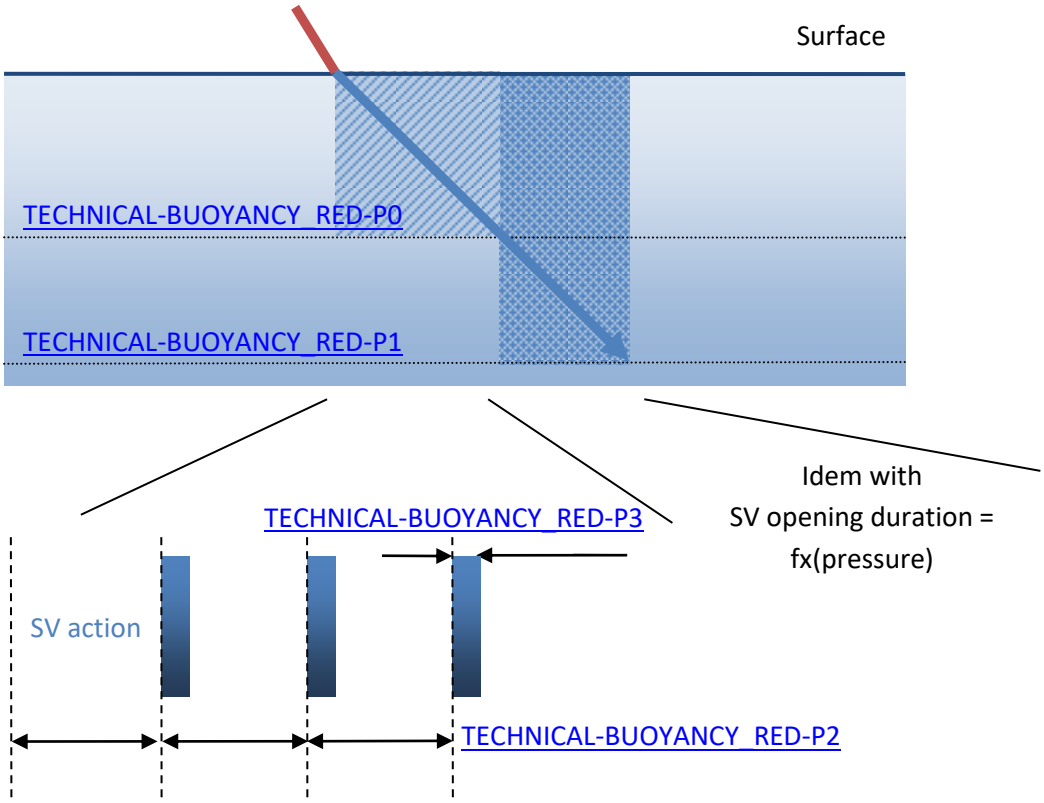
Two modes are proposed:

- “SECURE” mode
- “FAST” mode

Secure mode

The operation is as follows:

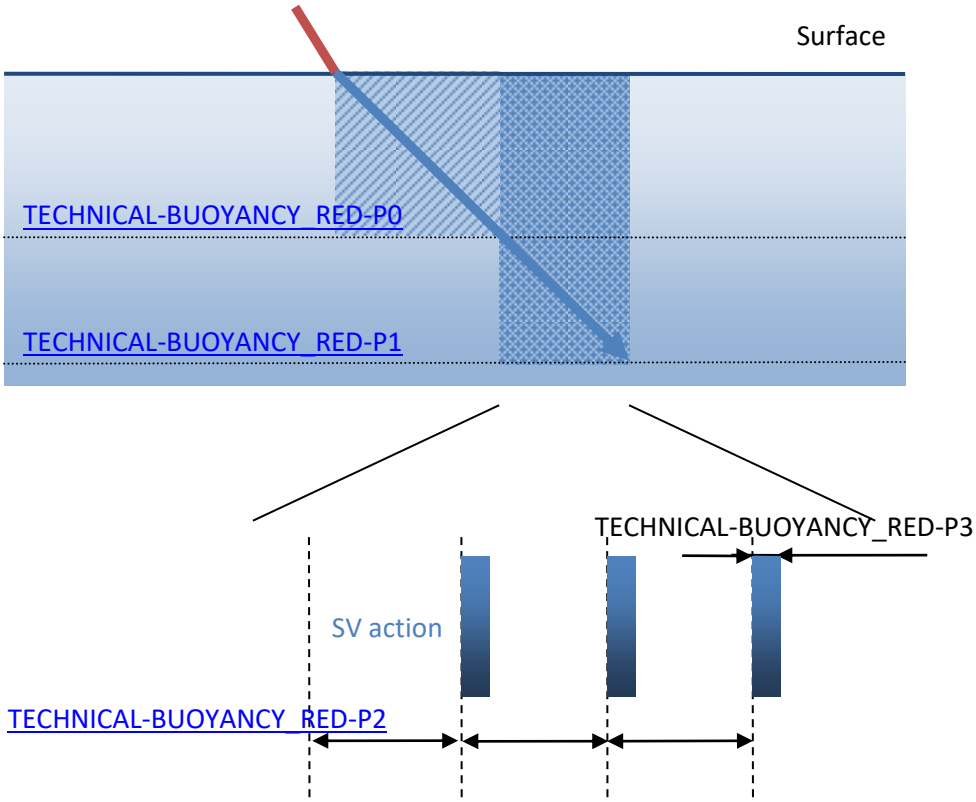
- Emergence reduction in two stages:
  - “Lump sum” phase corresponding to the transfer of the ¾ of the previous volume expected to sink (permanent SV opening)
  - “Complementary” phase respecting a waiting delay between each new activation to reach:
    - Threshold 1 by a standard opening duration of SV
    - Threshold 2 by a modulated opening duration of SV
- During this phase, only fast pressure request permits to check the thresholds.



Fast mode

The operation is as follows:

- Emergence reduction in two stages:
  - "Quick" phase up to threshold 1 (permanent SV opening)
  - "Standard" phase up to threshold 2 (modulated SV opening)
- During both phases, pressure monitoring is active (active CTD mode). If the threshold is exceeded, the SV is closed immediately.
- The modulation of SV opening in the "standard" phase consists in performing SV activations (variable duration) spaced from one another (the shorter and more spaced SV actions are, the more precise the detection of sinking will be).



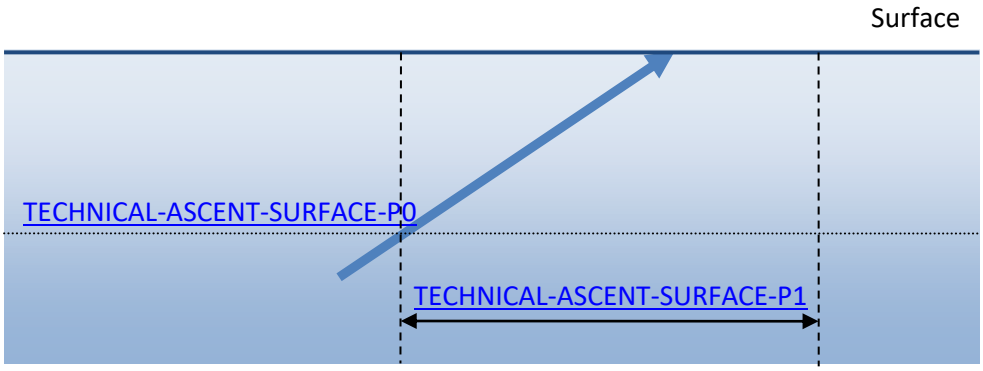
Before initiating the emergence reduction, the system performs the external pressure offset. This stage may be conditioned by the success of at least one of the surface actions.

5.2.Awaiting the end of ascent

Awaiting the end of ascent consists in allowing the ascending float to continue its course for few minutes following the detection of the approaching surface.

During this waiting period:

- No hydraulic action takes place
- Sensors are still in acquisition mode

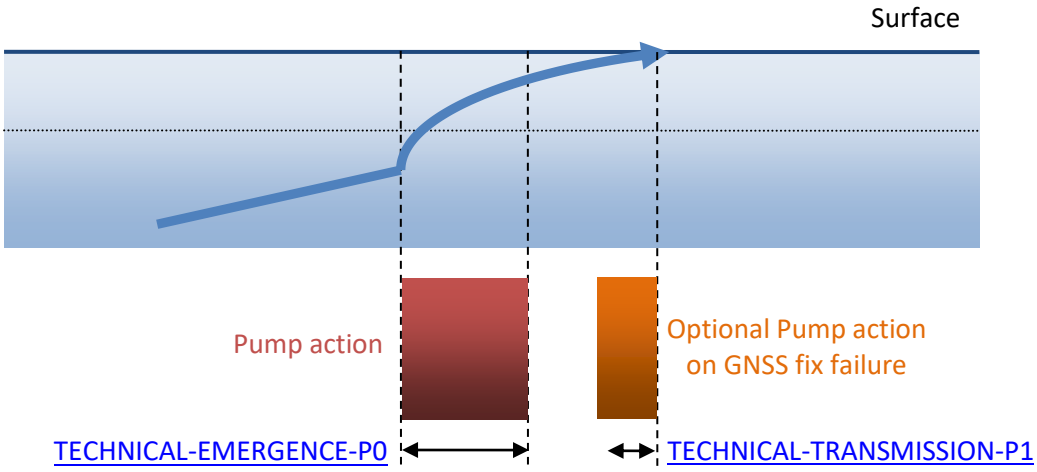


5.3.Emergence

The emergence consists in providing sufficient buoyancy to the float to perform a session at the surface.

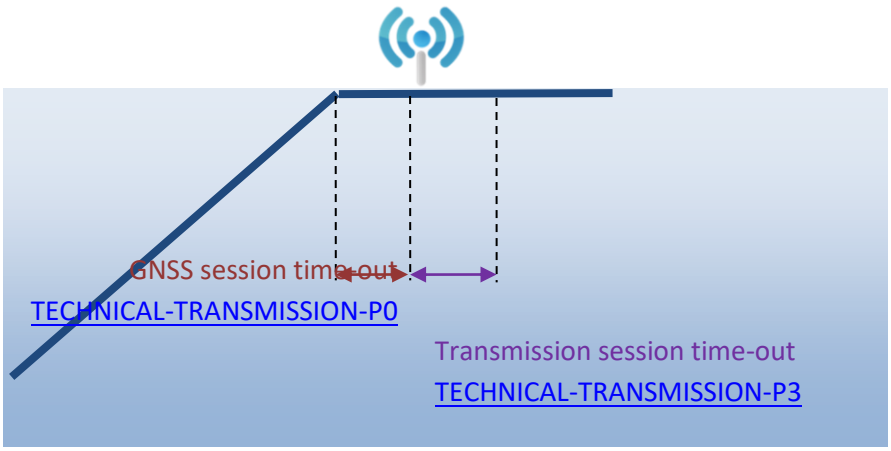
The operation is as follows:

- A fixed pump action takes place during the time spent at the surface
- A second action takes place if a failure occurs on acquisition of GNSS valid fix

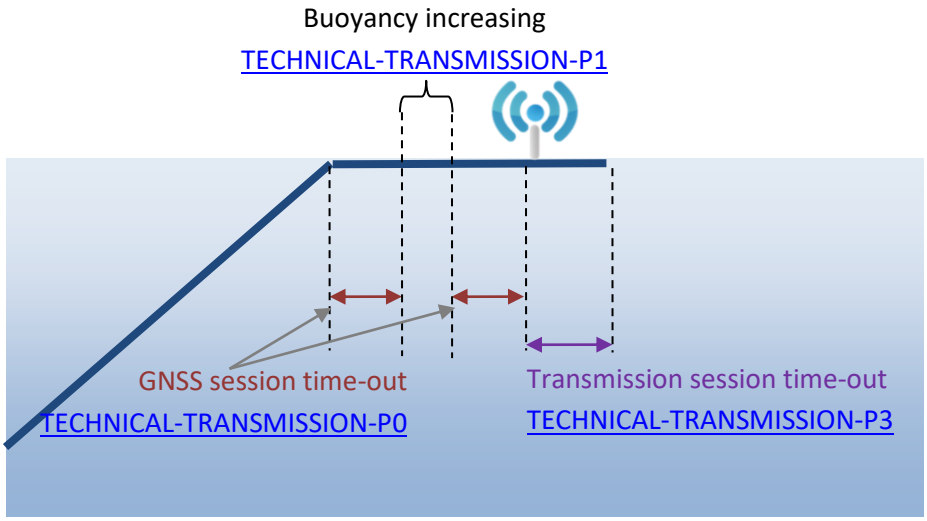


5.4.Transmission

The transmission phase consists of acquiring a geographical position and then transmitting all the data from the cycle by satellite, along with any previous cycles that have not been transmitted.



--- Figure 1. Typical transmission session ---



--- Figure 2. Transmission session with GNSS failure ---

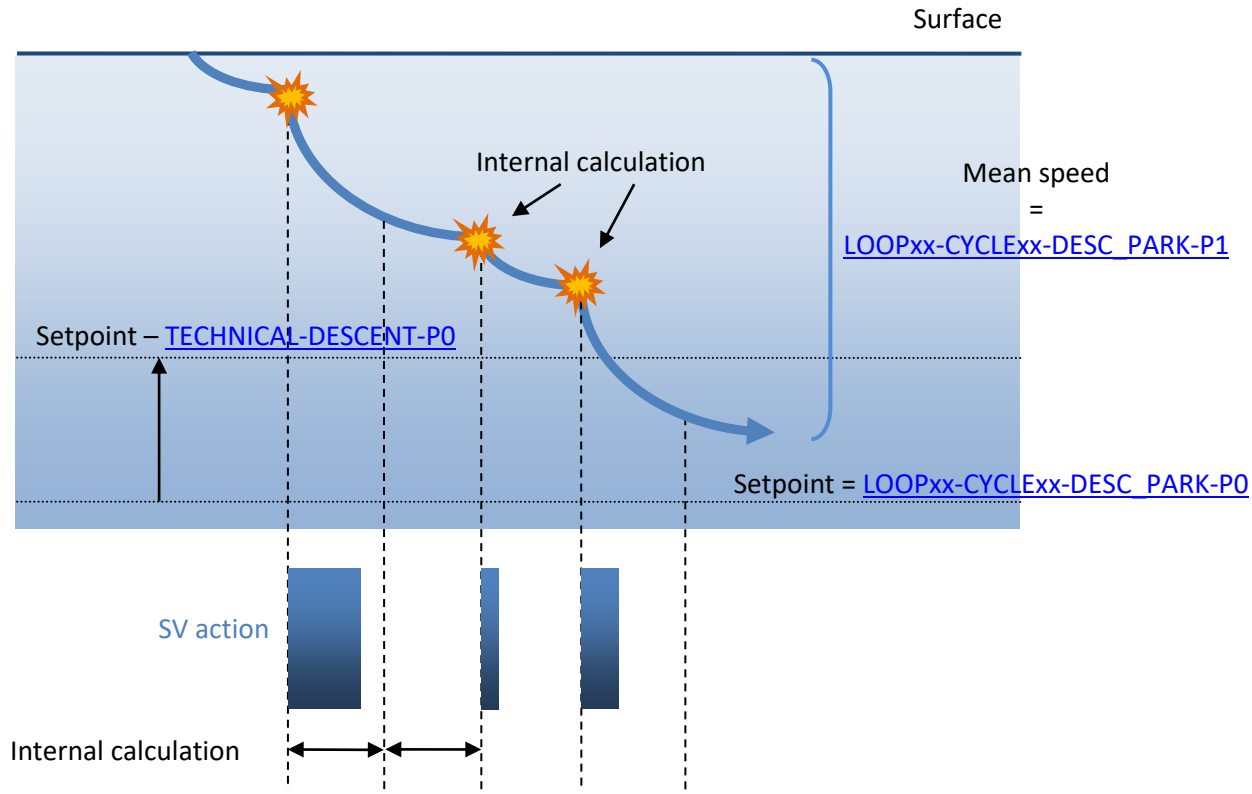
6. Navigation

6.1.Descent

The descent consists in reaching a set depth by respecting the expected mean speed while controlling the navigation speed of the float.

The operation is as follows:

- Rate of descent monitoring at fixed time intervals
- SV actions modulated over time
- End of descent when approaching the set point



Same strategy applied for the “descent to profile” phase by using the parameters LOOPxx-CYCLExx-DESC_PROF-P0 and LOOPxx-CYCLExx-DESC_PROF-P1.

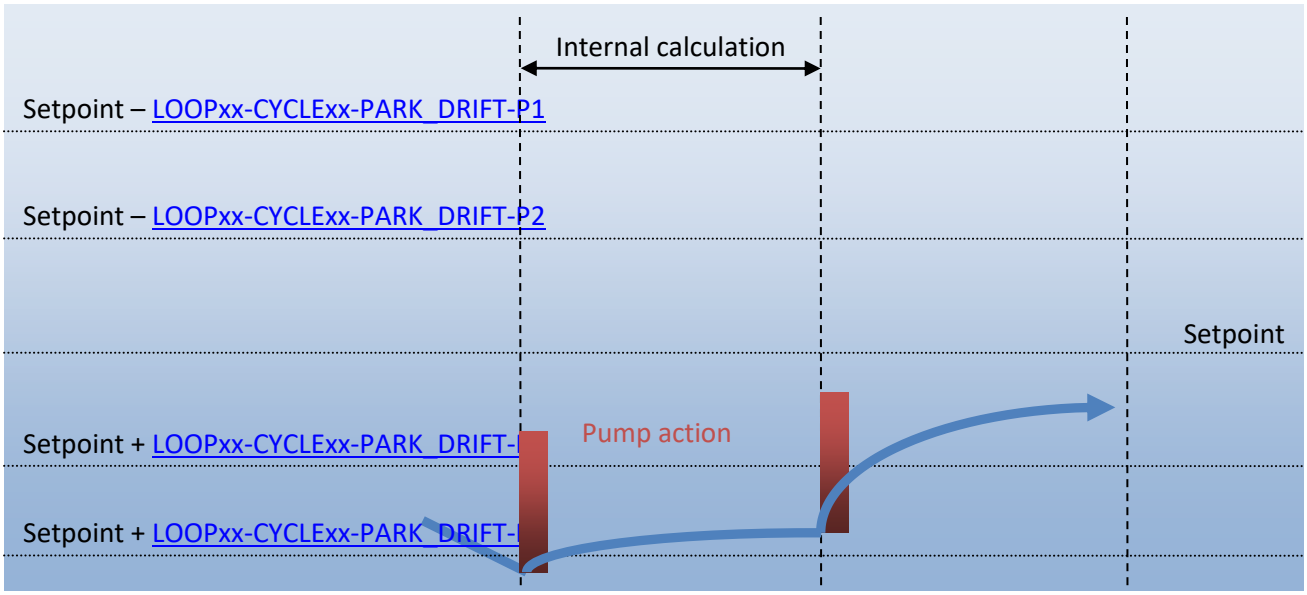
6.2.Repositioning during drift

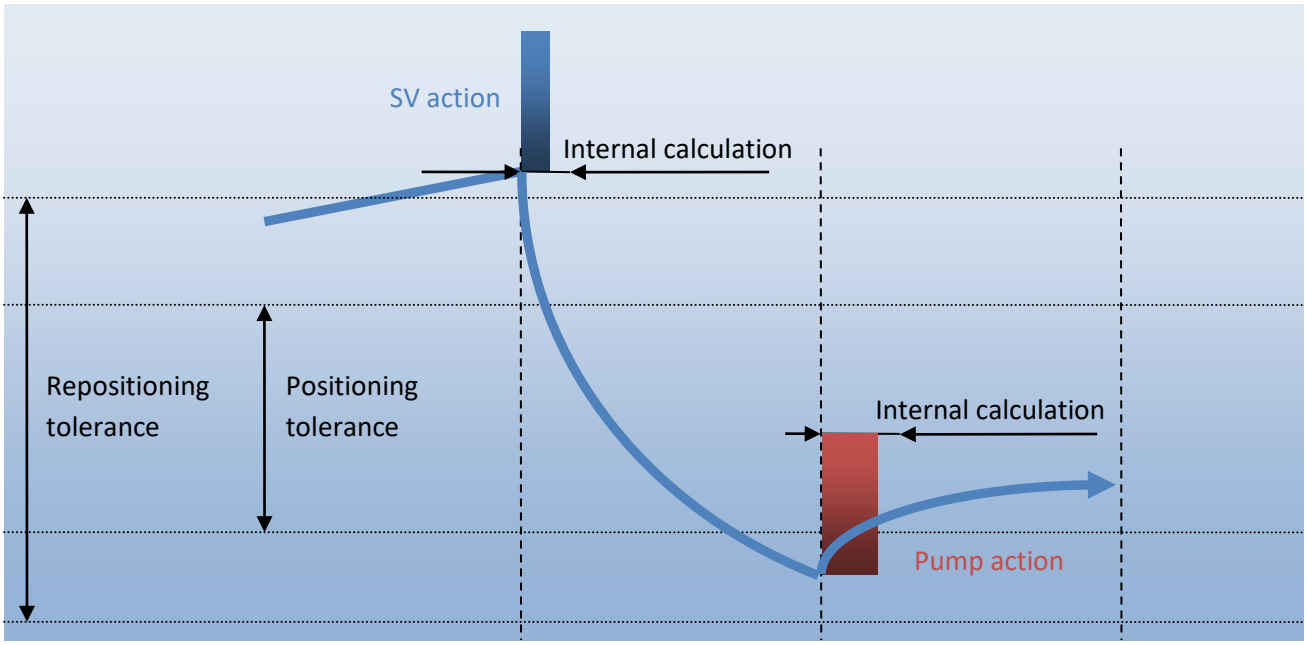
Repositioning during drift consists in maintaining the float around a set depth.

The operation is as follows:

- Pressure monitoring at fixed time intervals
- Repositioning if the current pressure deviates from the set point by more than the repositioning tolerance

- Repositioning completed when the current position is deviated from the set point by less than the positioning tolerance



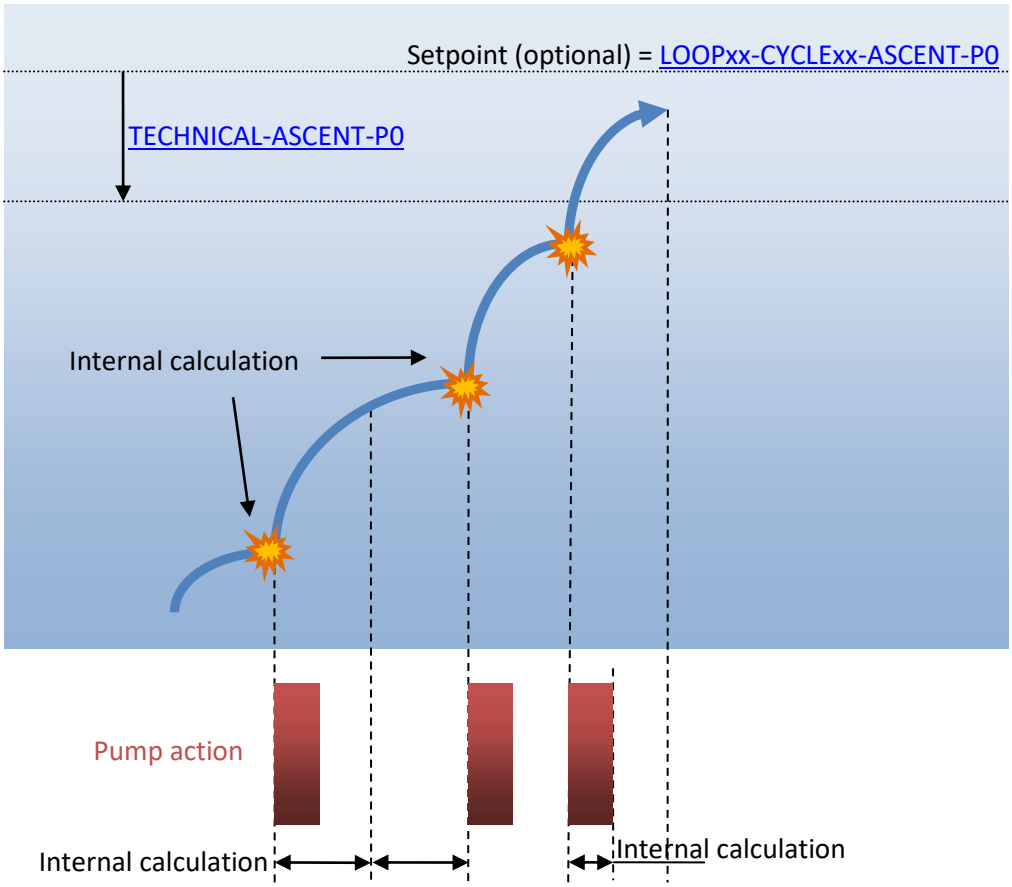


6.3.Ascent

The ascent consists in reaching a set depth while controlling the navigation speed of the float.

The operation is as follows:

- Rate of ascent monitoring at fixed time intervals
- Fixed pump actions
- End of ascent when approaching the set point



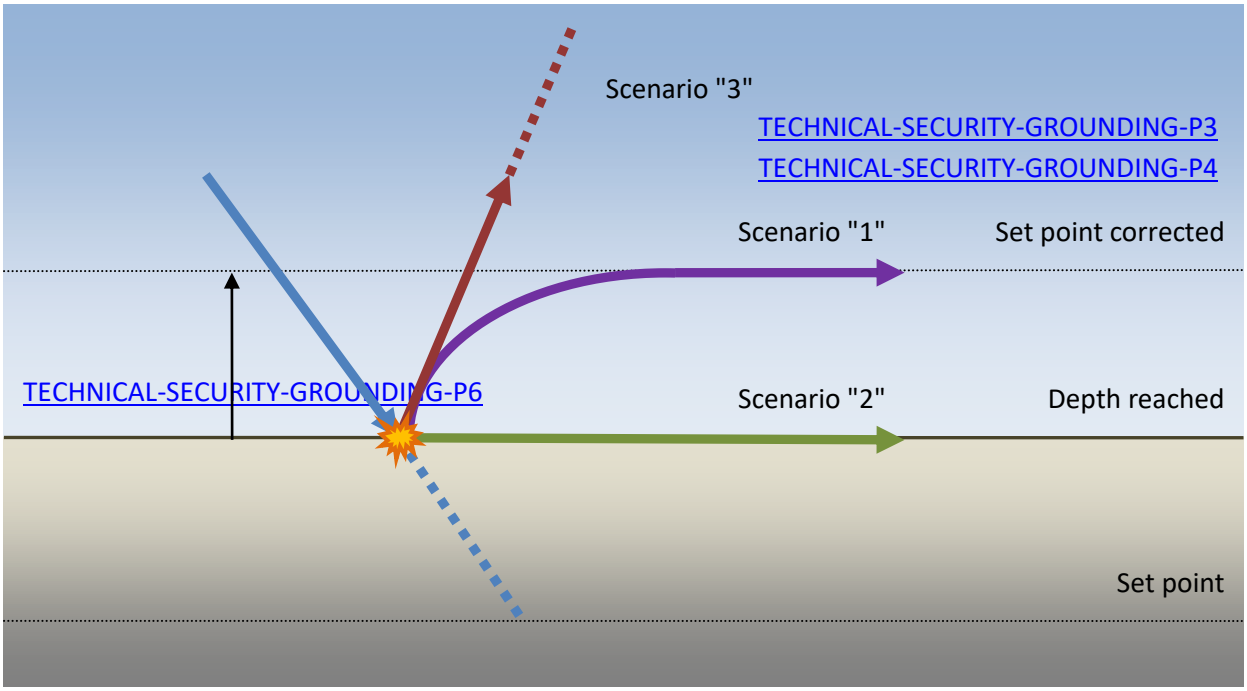
7. Management of degraded modes

7.1.Grounding

The grounding during the descent consists in detecting the absence of movement of the float despite a large displacement of oil to descend.

The operation is as follows:

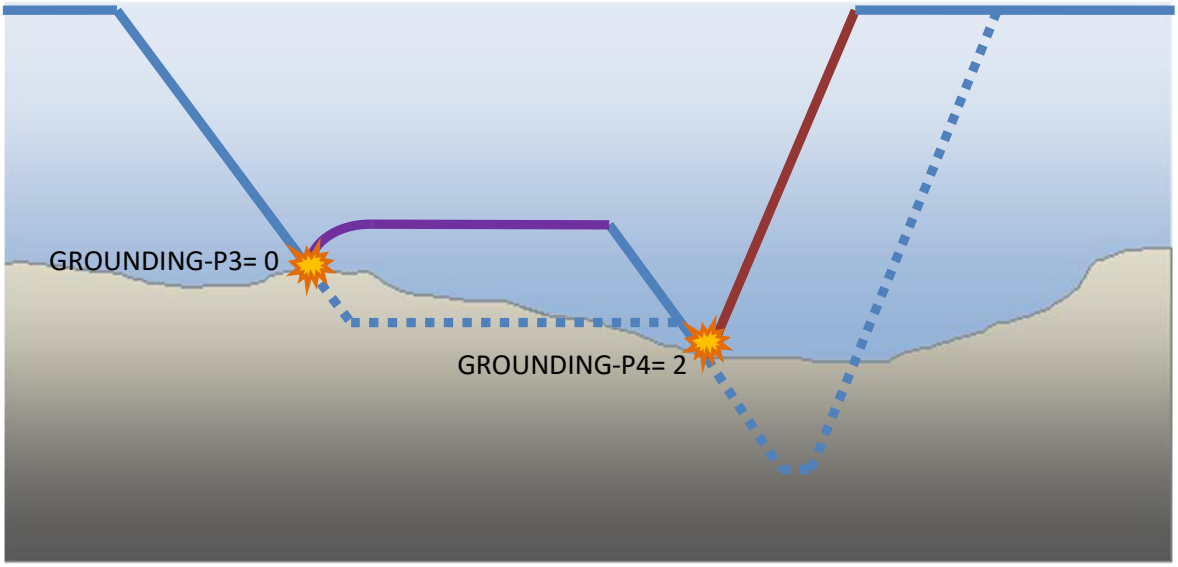
- A safety task monitors the travel speed of the float and detects stabilisations less than TECHNICAL-SECURITY-GROUNDING-P2
- When stabilisation is detected, the cumulative volume of oil is compared with a threshold TECHNICAL-SECURITY-GROUNDING- P1
- Upon detection of grounding:
    - **Scenario "1": Escape mode - Set point correction**
      The pre-programmed drift depth is disregarded. The pressure at the time of grounding minus an offset (TECHNICAL-SECURITY-GROUNDING-P6) is taken as the new value for the drift pressure. The float adjusts its buoyancy to reach this new drift depth. The drift depth stays unchanged for the subsequent cycles. If the grounding pressure is lower than the programmed threshold (TECHNICAL-SECURITY-GROUNDING-P5), the float remains on the seabed until the ascent time.
    - **Scenario "2": Static mode – Stay on seabed**
      The float remains where it is until the ascent time. The pressure at the time of grounding becomes the ascent profile start pressure for the current cycle. The ascent profile start pressure stays unchanged for the subsequent cycles.
    - **Scenario "3": Early surface mode – Ascent immediately**
      The float starts to ascent immediately to reach the surface.



The strategy to adopt in case of grounding is configured for descent to parking depth and parking drift stages by TECHNICAL-SECURITY-GROUNDING-P3.
The strategy to adopt in case of grounding is configured for descent to profile depth and profile depth drift stages by TECHNICAL-SECURITY-GROUNDING-P4.

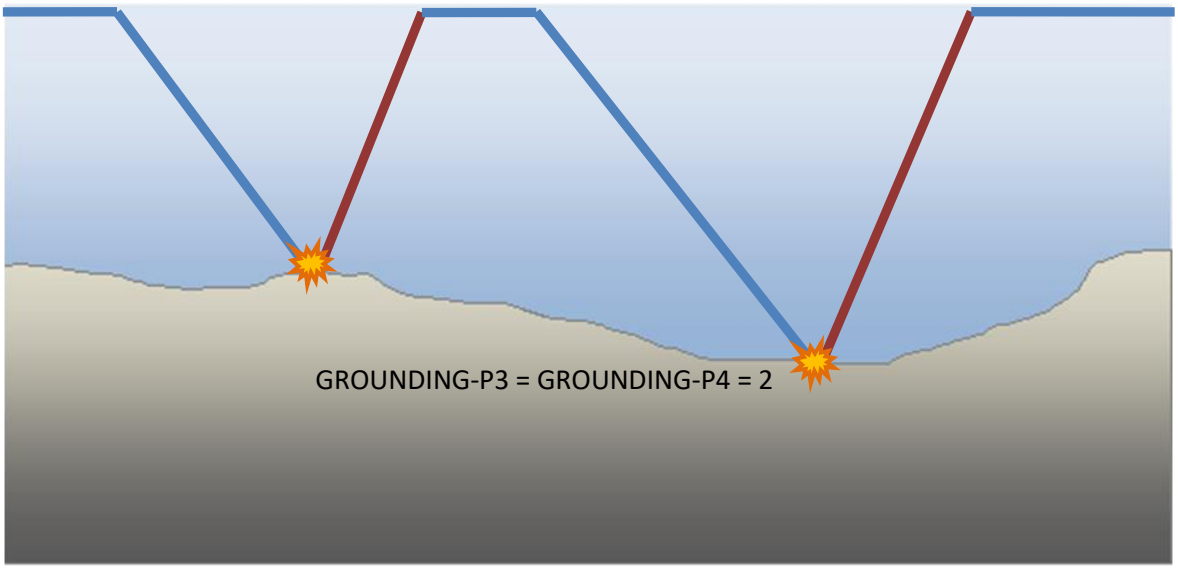
7.1.1. Scenario 1: Escape mode – Setpoint correction

In this configuration, the profiler corrects its parking depth to continue to dive during parking drift stage.



7.1.2. Scenario 3: Early surface mode – Ascent immediately

In this configuration, the profiler stops the descent stage and start the ascent stage when encountering the ground.

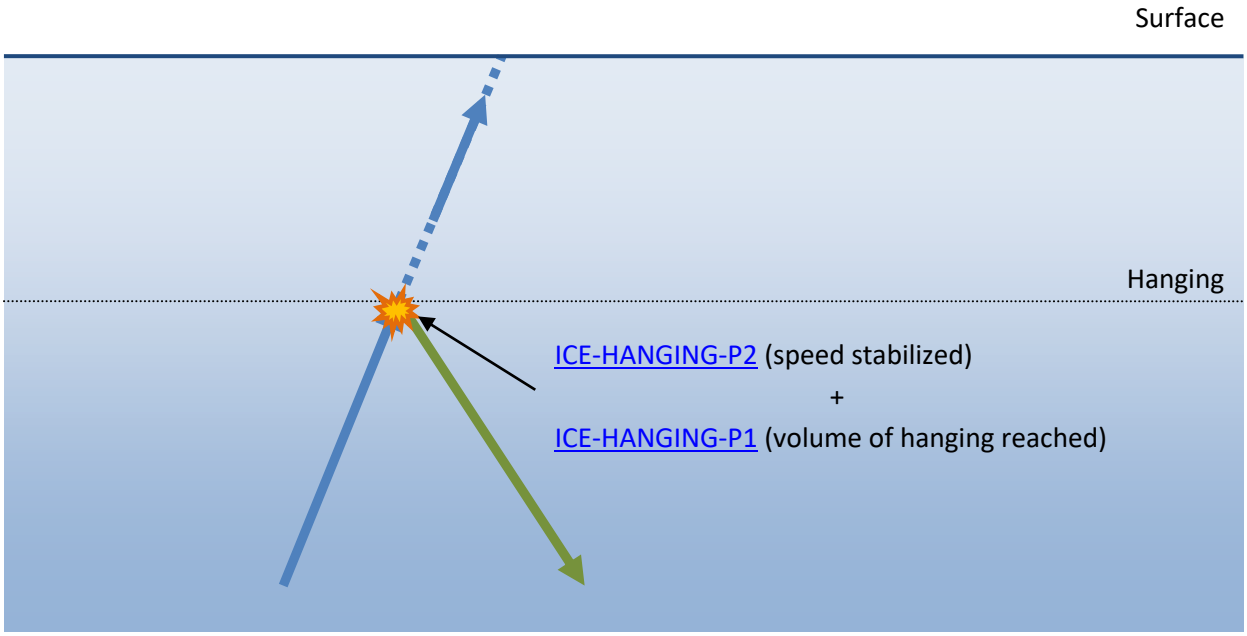


7.2.Hanging (only if “Ice detection” enabled)

The hanging during the ascent consists in detecting the absence of movement of the float despite a large displacement of oil to ascend.

The operation is as follows:

- A safety task monitors the travel speed of the float and detects stabilisations less than ICE-HANGING-P2
- When stabilisation is detected, the cumulative volume of oil is compared with a threshold ICE-HANGING-P1
- Upon detection of hanging during ascent, the navigation continues without performing any emergence.



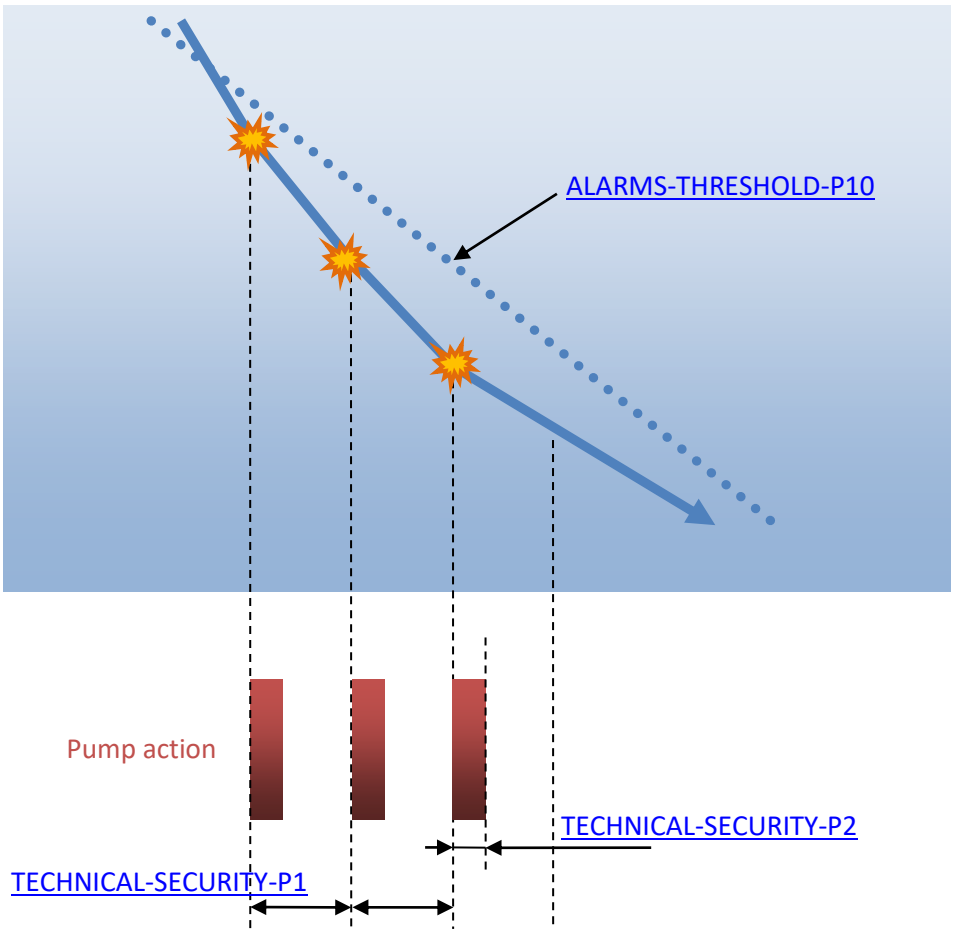
Note: When the hanging is not activated, the float will abort the ascent after few hours (ex: 48 hours for a theorical ascent delay for 9cm/s with a coefficient of security of 8).

7.3.Braking during descent

Braking during descent consists in performing pump actions to slow down the sinking of the float if the speed limit is exceeded.

The operation is as follows:

- A safety task monitors the travel speed of the float and detects high speeds
- In case of abnormal speed, a fixed pump action takes place

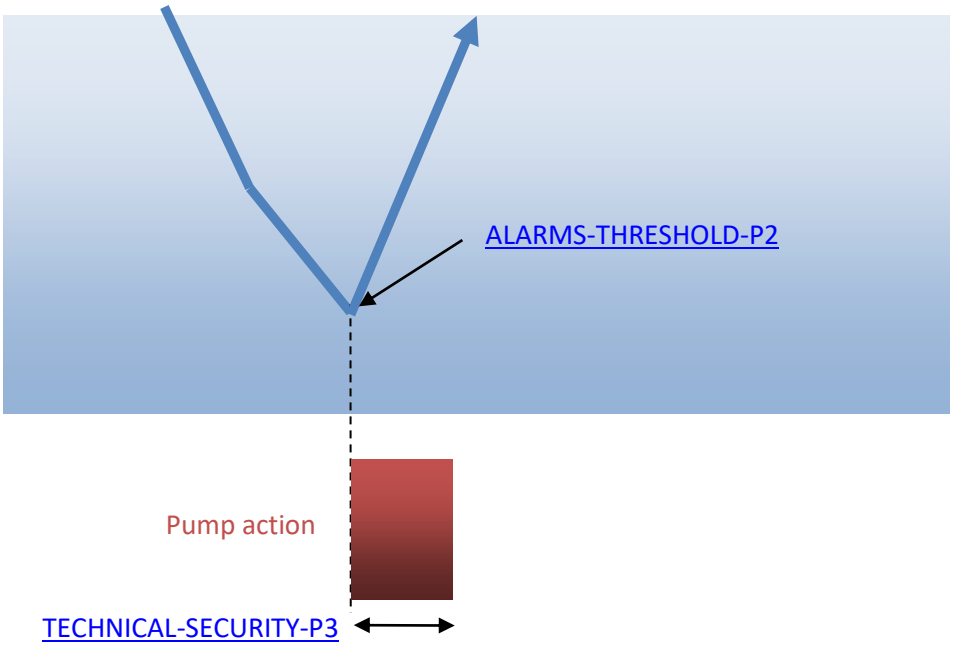


7.4.Braking on exceeding max. pressure

This braking on descent consists of a rapid pumping action to stop the float exceeding its maximum pressure before starting an emergency ascent.

The operation is as follows:

- A safety task monitors the travel depth of the float and checks the pressure.
- In case of the maximum pressure reached, a fixed pump action takes place and the float switches into end of life mode.



8. End of life

End of life can occur due to:

- At the end of the programmed maximum number of cycles
- In case of serious technical issue (user selectable event)
- Forced by remote control

The events that will trigger this procedure can be chosen from a list.

The (optional) alarms that can trigger this procedure are as follows:

- Low battery voltage
- High external pressure
- High internal pressure
- High descent speed
- External pressure no longer responding
- Detection of water inside the float

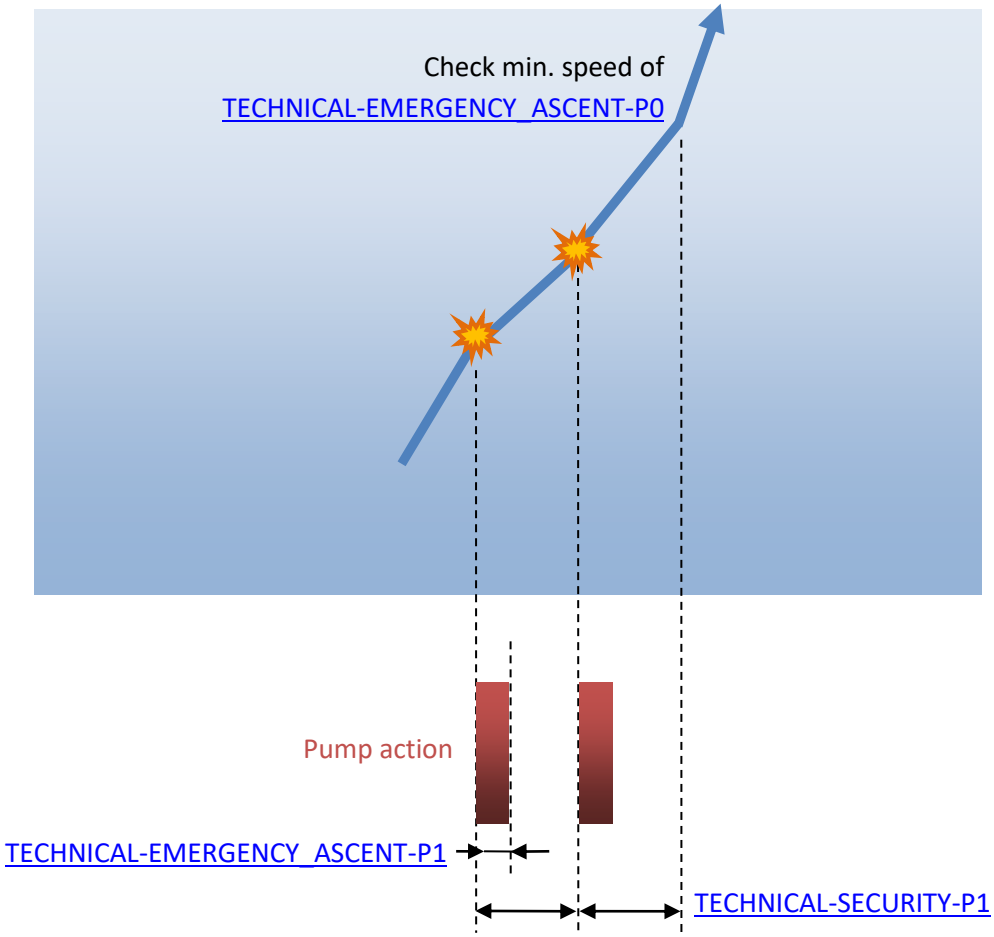
The (not modifiable) alarms that can trigger this procedure are as follows:

- Invalid CTD setting
- RTC error

8.1."Emergency" ascent

The "Emergency" ascent consists in bringing the float back to the surface before switching to the end of life mode.

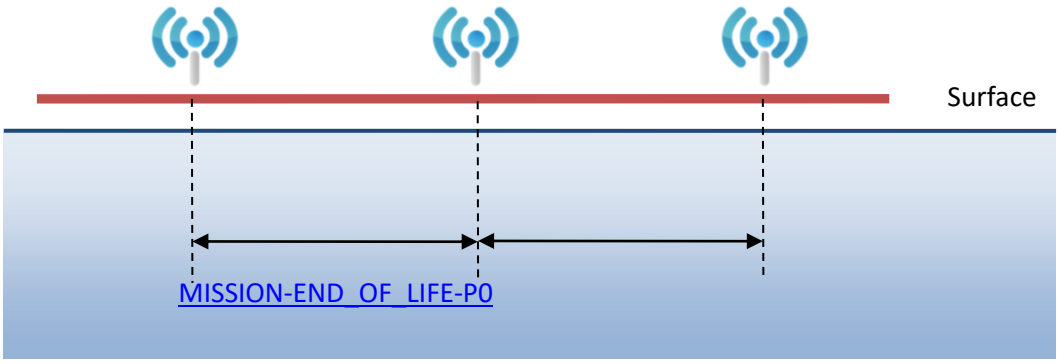
This ascent is called "emergency" because it implies that a critical event occurred while the float was in navigation mode. The float is then placed in maximum buoyancy state to reach the surface as quickly as possible.



8.2."End of life" mode

"End of Life" mode consists of acquiring the GPS position (optional) and transmitting it at regular intervals. In this phase, the float remains on the surface and reports its status with its last geographical position. This mode is also designed to facilitate recovery of the float in the

ocean.



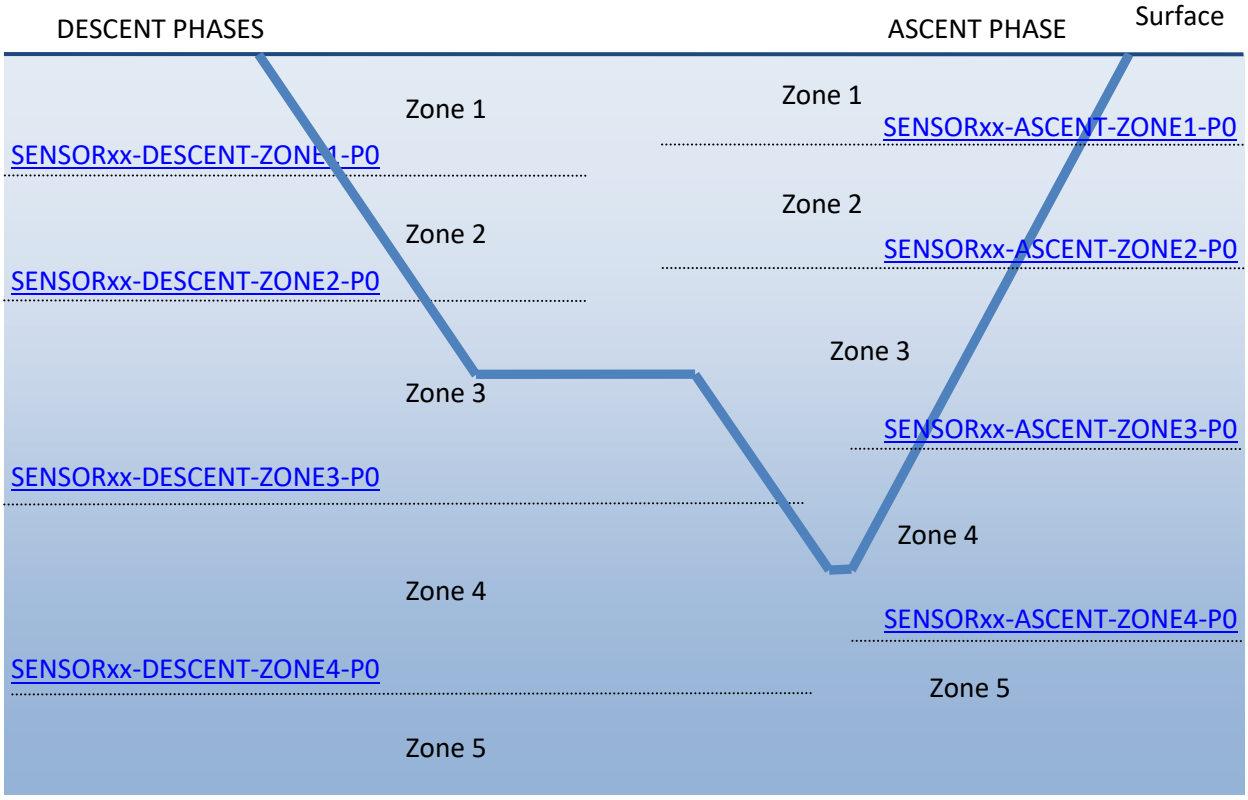
9. Standard sensors

Measurements of standard parameters can be done during descent, drift, ascent and surface phases. Each sensor is managed independently:

- Status (enabled/disabled)
- Zone separation
- Acquisition rate (per zone and per phase)
- Control type (per zone)
- Processing type (per zone)
- Section depth (per zone)
- Specific parameters

9.1. Concept of zone

The water column is divided into 5 different depth intervals to adapt the management of sensors to the experiment. These 5 zones are independent between descent and ascent. And each depth interval (zone 1 to 5) is also managed independently.



9.2.Power supply control

The type of power supply control for each sensor is managed independently according to the zone SENSORxx-DESCENT/ASCENT-ZONEx-P2 :

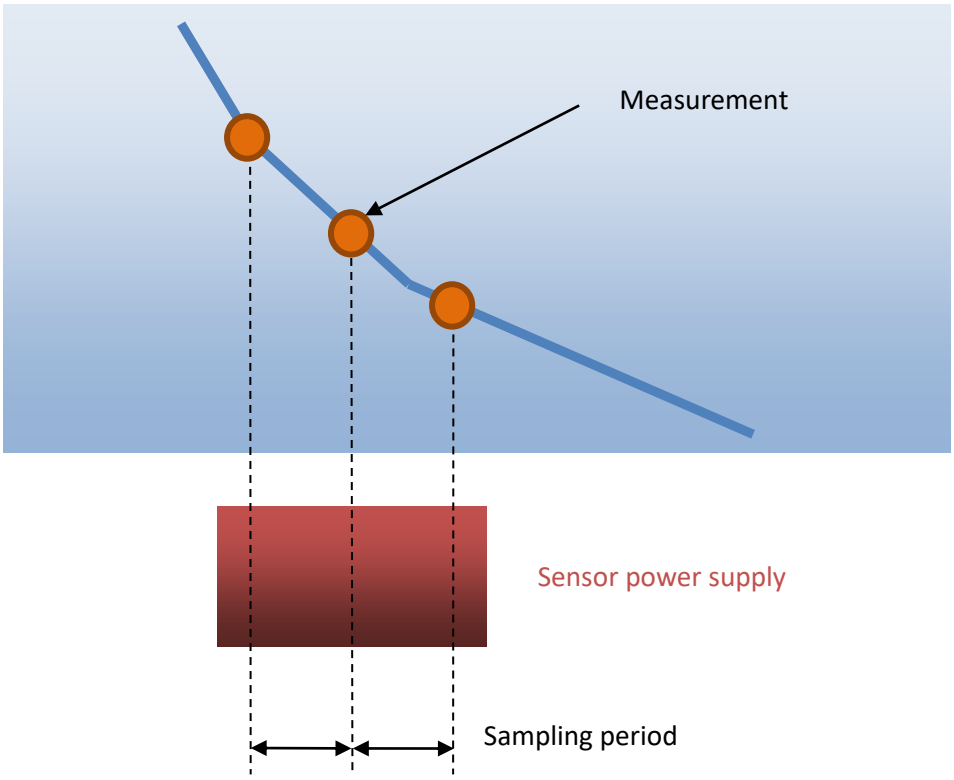
- None (0)
- Continuous (1)
- Pulsed (2)
- Spot sampling (3)

9.2.1. None (0)

In "none" mode, there is no power supply control. The sensor is switched off in the zone.

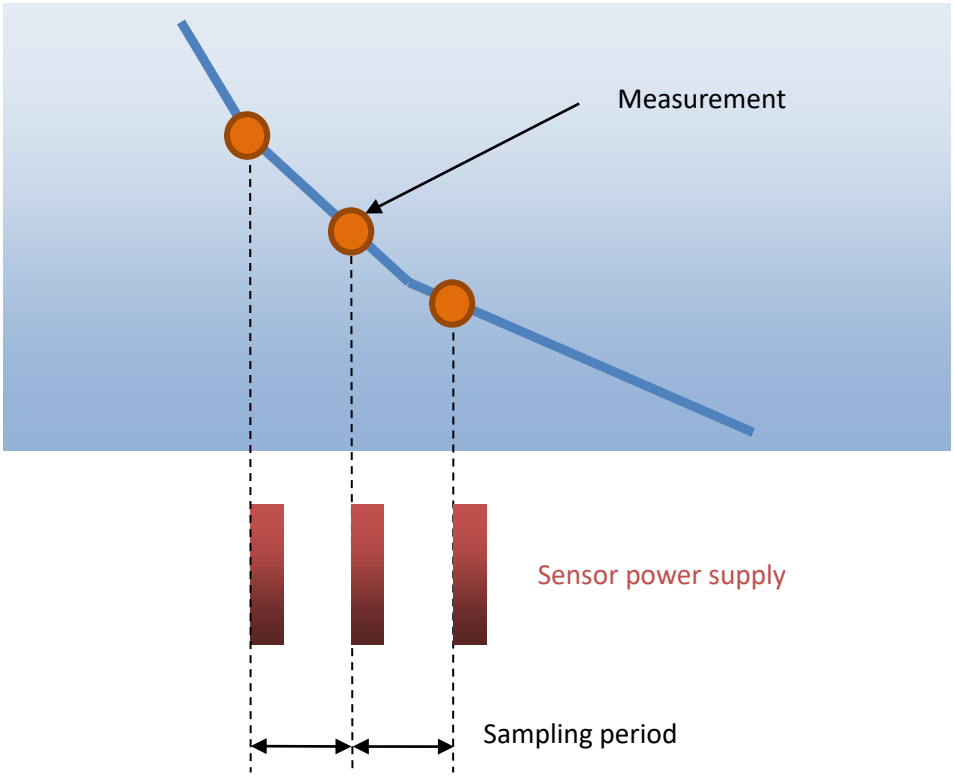
9.2.2 Continuous (1)

In "continuous" mode, the sensor is powered continuously.



9.2.3 Pulsed (2)

In "pulsed" mode, the sensor is powered at regular intervals according to the acquisition rate. The sensor is switched off after the measurement.



9.2.4 Spot sampling (3)

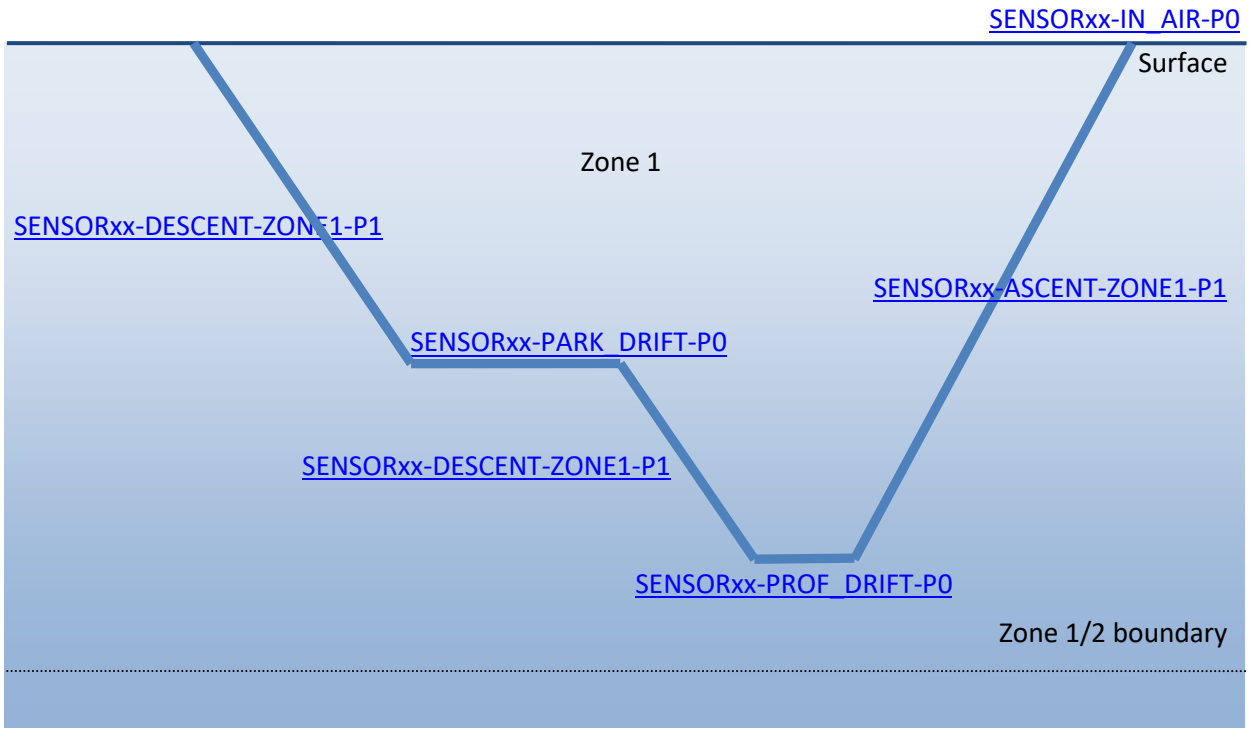
In “spot sampling” mode, the sensor is powered just before the middle of the expected slice thickness. The sensor acquires few samples (SENSORxx-DESCENT/ASCENT-ZONEx-P5) and then is powered off.

9.3 Acquisition rate

The acquisition rates of each sensor are managed independently according to the navigation phase:

- Descent to "parking" drift depth
- Drift at "parking" depth
- Descent to "profil" depth
- Drift at "profil" depth
- Ascent to surface
- Surface for “in air measurement”

It is possible to disable the acquisition in the phase by setting the rate at "0" seconds.



Note: Parameters not applicable if the "Spot sampling" mode is selected.

9.4 Data processing

The data of each sensor can be processed independently according to each zone SENSORxx-DESCENT/ASCENT-ZONEx-P3:

- Raw (0x00)
- Raw decimated (0x80)
- **Arithmetic mean decimated (0x81) → Usual configuration**

Note: Some restrictions on data processing are implemented on the float:

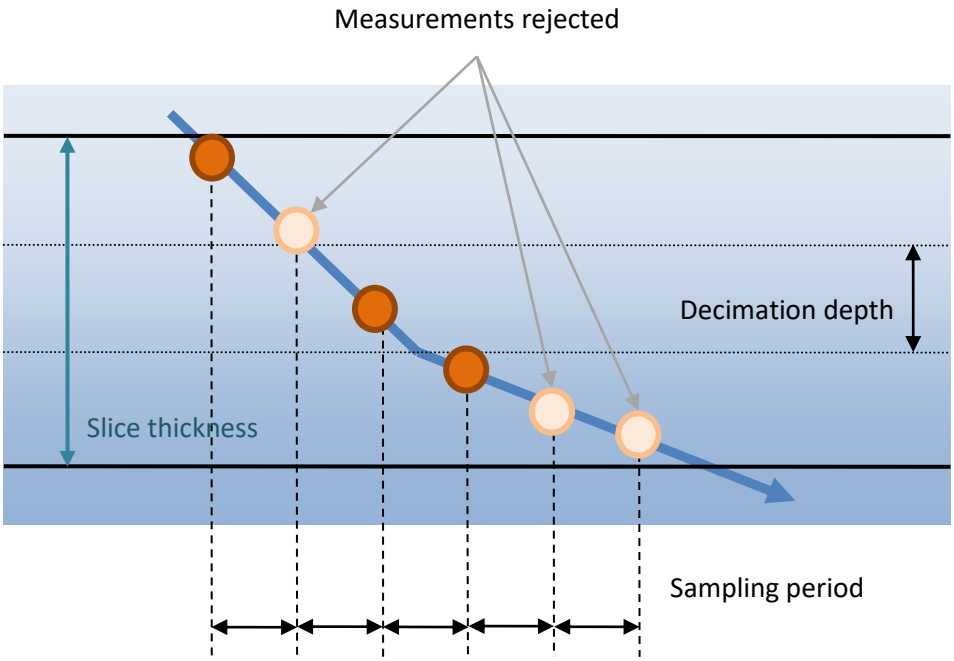
- During descent and ascent stages all data processing are available
- During drift stages data processing is fixed to raw processing
- During surface stage data processing is fixed to raw processing

9.4.2 Decimation of measurements

When a type of processing is configured, the various points acquired can be decimated (according to pressure) so as not to influence the processing (in case the points are badly distributed over the water column).

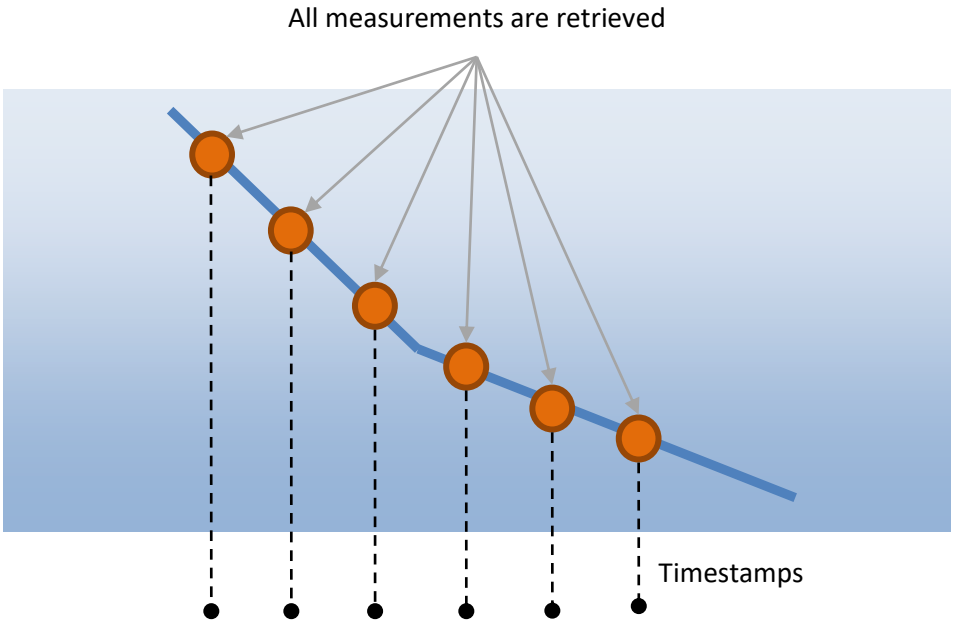
Decimation consists in keeping a single point per decimation depth. Decimation depth is selected automatically by the system according to the slice thickness SENSORxx-DESCENT/ASCENT-ZONEx-P4:

- Depth of 0.1 dbar for slice thickness of 1 dbar and less
- Depth of 0.2 dbar for slice thickness between 1 and 5 dbar
- Depth of 0.5 dbar for slice thickness of 5 dbar or more



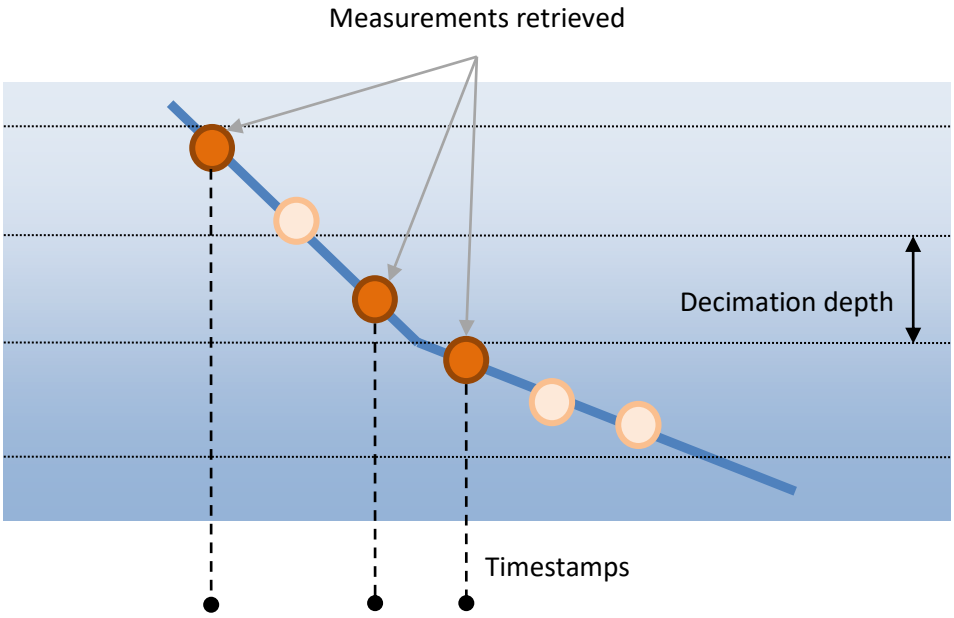
9.4.3 Raw (0x00)

In "raw" mode, the acquired data are retrieved without decimation.



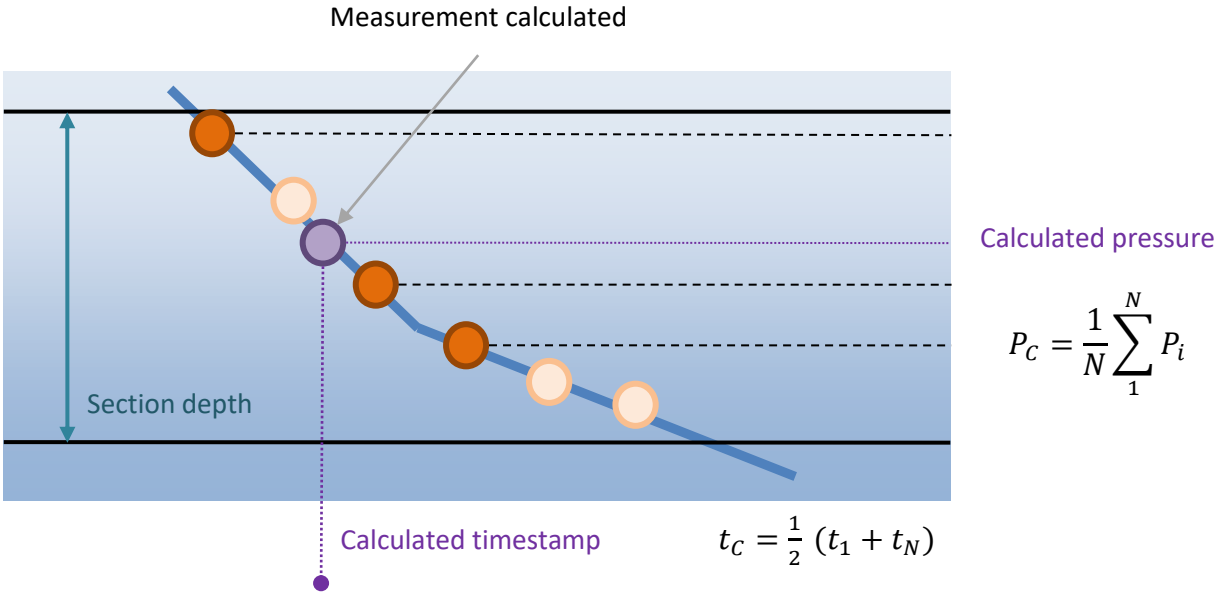
9.4.4 Raw decimated (0x80)

In “raw decimated” mode, the acquired data are decimated according to the section depth. In this case, decimation depth is equal to the slice thickness.
The lowest slice thickness is 0.1 dbar.



9.4.5 Arithmetic mean decimated (0x81)

In "average" mode, the decimated data for the section are averaged and the calculated measurement is time-stamped. The lowest slice thickness is 0.1 dbar.

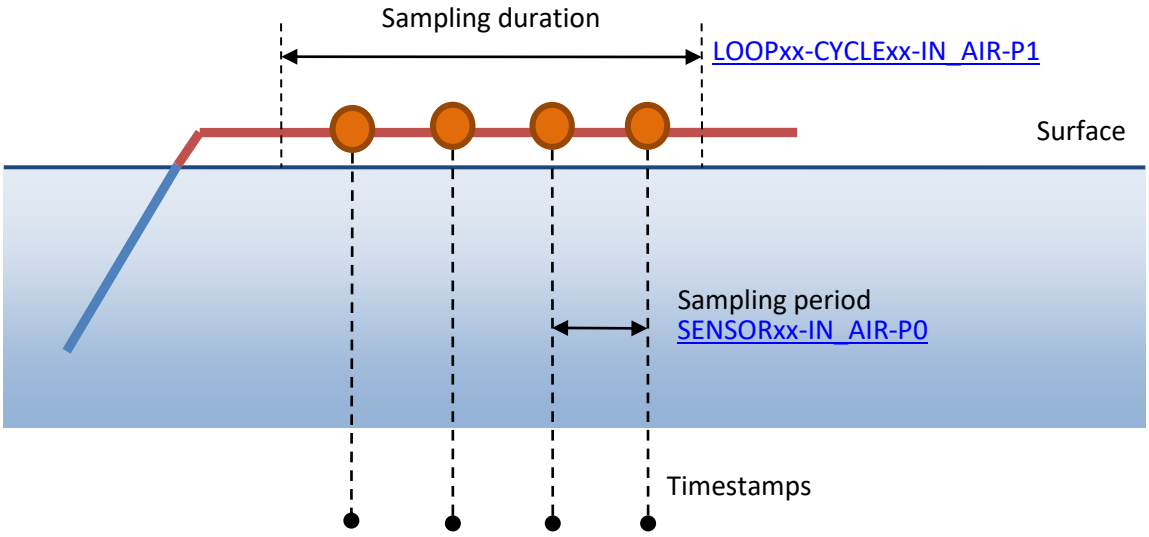


9.5 Surface acquisition

In-air measurements can be achieved during the surface stage after the float had inflated its external bladder. This feature allows post correction of sensor's data.

This function can be activated by sensor and the surface sampling period set independently by sensor while the total acquisition time is common to all sensors.
The phase is not mandatory in every cycle. It is possible to create a cycle frequency for the execution of this phase:
Cycle periodicity = 1 / LOOPxx-CYCLExx-IN_AIR-P0.

- P0 = 0 ➔ No "in air" phase
- P0 = 1 ➔ "In air" phase on each cycle
- P0 = x ➔ "In air" phase every "x" cycles

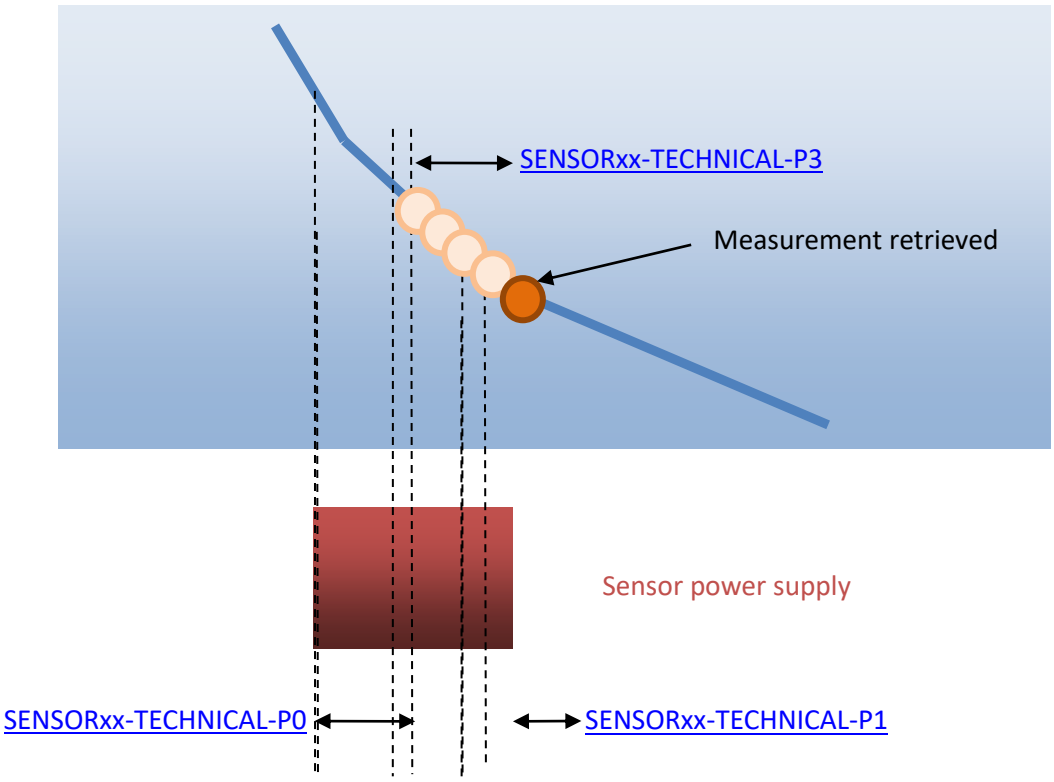


Note: During surface acquisition stage, the power supply control mode is defined by SENSORxx-IN_AIR-P1.

9.6 Common specific parameters

The specific parameters of each sensor are managed independently according to the set of standard parameters:

- Warm-up time required by the sensor before taking measurements
- Shut down time required for the sensor to stop
- Measurement filtering index before considering that the measurements are correct



9.7 Identification of sensors

Following sensors can be embedded on float:

- SENSOR01: SBE41
- SENSOR02: Aanderaa4330
- SENSOR03: RBRargo3

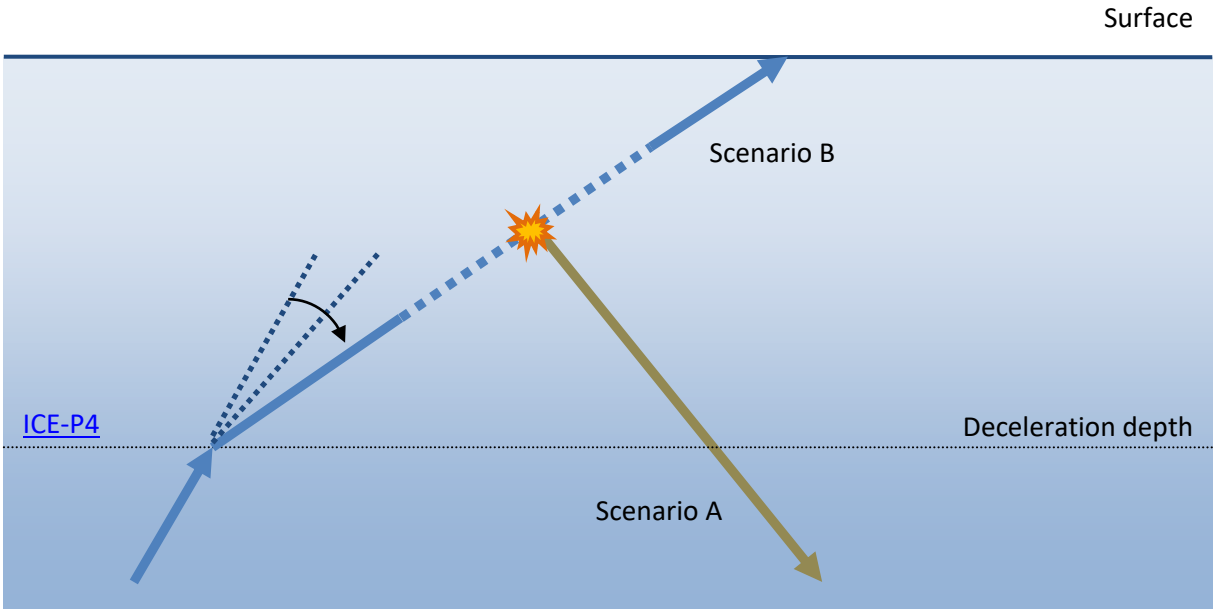
10 Extended features

10.1 Ascent with "ice detection"

This type of pattern consists in dividing the ascent in different stages to allow the "measurement" to detect the presence of ice and to abort the pattern/cycle.

The operation is as follows:

- Stage 1: "Standard" ascent to the deceleration depth
- Stage 2: "Slow" ascent to ice detection test depth
- Stage 3: Waiting for ice detection
- Stage 4: Two possibilities
    - Scenario A: abort by the "ice detection" and switch to the next cycle
    - Scenario B: the ascent resumes



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